STA 445 S26 Assignment 6

2026-04-03

Load your packages here:

Problem 1

A common task is to take a set of data that has multiple categorical variables and create a table of the number of cases for each combination. An introductory statistics textbook contains a data set summarizing student surveys from several sections of an intro class. The two variables of interest are `Gender` and `Year` which are the students gender and year in college. *Note: you will need to refer to Chapter 4 and Chapter 7 for some of the operations needed below - this is a great time to review chapter 4!*

a) Download the data set using the following:

```
Survey <- read.csv('https://www.lock5stat.com/datasets2e/StudentSurvey.csv', na.strings=c('',' ')
```

b) Select the specific columns of interest **Year** and **Gender**

```
COI <- Survey %>%
  select(Year, Gender)
COI
```

```
##          Year Gender
## 1      Senior      M
## 2   Sophomore      F
## 3   FirstYear      M
## 4      Junior      M
## 5   Sophomore      F
## 6   Sophomore      F
## 7   FirstYear      F
## 8   Sophomore      M
## 9      Junior      F
## 10  FirstYear      F
## 11  Sophomore      F
## 12  FirstYear      M
## 13  Sophomore      M
## 14     Junior      F
## 15  FirstYear      M
## 16  FirstYear      F
## 17  FirstYear      F
## 18  Sophomore      M
```

```
## 19  Sophomore      F
## 20  Sophomore      F
## 21  Sophomore      F
## 22  Sophomore      F
## 23  FirstYear      M
## 24  Sophomore      M
## 25  Sophomore      M
## 26  Sophomore      M
## 27  Sophomore      F
## 28     Junior      M
## 29  FirstYear      M
## 30     Senior      M
## 31     Junior      M
## 32  Sophomore      F
## 33  FirstYear      M
## 34     Senior      M
## 35  FirstYear      F
## 36  Sophomore      F
## 37  FirstYear      M
## 38  Sophomore      F
## 39  Sophomore      F
## 40     Junior      F
## 41     Senior      M
## 42  Sophomore      F
## 43  Sophomore      F
## 44  FirstYear      F
## 45  FirstYear      M
## 46     Junior      M
## 47     Senior      M
## 48  Sophomore      M
## 49     Senior      M
## 50  Sophomore      F
## 51  FirstYear      M
## 52     Senior      F
## 53     Junior      M
## 54     Senior      M
## 55     Junior      M
## 56  FirstYear      M
## 57     Junior      F
## 58  Sophomore      F
## 59  Sophomore      F
## 60  FirstYear      F
## 61  Sophomore      M
## 62     Senior      M
## 63  FirstYear      F
## 64  FirstYear      M
## 65  FirstYear      M
## 66  Sophomore      F
## 67  FirstYear      M
## 68  Sophomore      M
```

```
## 69  Sophomore      M
## 70     Senior      M
## 71     Senior      F
## 72  Sophomore      F
## 73  FirstYear      F
## 74  FirstYear      F
## 75  Sophomore      F
## 76  FirstYear      F
## 77  FirstYear      F
## 78  Sophomore      F
## 79  Sophomore      F
## 80  Sophomore      M
## 81  Sophomore      M
## 82  FirstYear      F
## 83  FirstYear      F
## 84     Junior      F
## 85  FirstYear      F
## 86  Sophomore      F
## 87     Senior      M
## 88  FirstYear      F
## 89  FirstYear      M
## 90  FirstYear      M
## 91  Sophomore      M
## 92  FirstYear      F
## 93  Sophomore      F
## 94  Sophomore      M
## 95  FirstYear      M
## 96  Sophomore      M
## 97  Sophomore      F
## 98  Sophomore      M
## 99  Sophomore      M
## 100 FirstYear      M
## 101    Senior      F
## 102 Sophomore      F
## 103 FirstYear      M
## 104 Sophomore      M
## 105 Sophomore      M
## 106      <NA>      F
## 107 Sophomore      M
## 108 Sophomore      M
## 109 Sophomore      F
## 110 Sophomore      M
## 111 Sophomore      M
## 112 Sophomore      M
## 113 Sophomore      F
## 114 FirstYear      M
## 115 FirstYear      F
## 116 Sophomore      F
## 117 Sophomore      F
## 118 Sophomore      M
```

```
## 119 Sophomore      M
## 120 FirstYear      F
## 121 Sophomore      F
## 122 FirstYear      M
## 123 Sophomore      M
## 124 Sophomore      M
## 125 Sophomore      F
## 126 Sophomore      M
## 127 Sophomore      M
## 128 FirstYear      M
## 129 Sophomore      F
## 130 Sophomore      M
## 131    Senior      F
## 132 FirstYear      F
## 133 Sophomore      F
## 134 Sophomore      F
## 135 Sophomore      F
## 136 Sophomore      F
## 137 Sophomore      M
## 138 FirstYear      F
## 139 Sophomore      F
## 140    Junior      F
## 141 FirstYear      M
## 142 FirstYear      F
## 143 FirstYear      M
## 144    Junior      M
## 145 FirstYear      F
## 146 Sophomore      F
## 147 FirstYear      F
## 148 Sophomore      M
## 149 Sophomore      M
## 150 FirstYear      F
## 151 Sophomore      F
## 152 FirstYear      M
## 153 FirstYear      F
## 154 Sophomore      M
## 155    Senior      M
## 156 Sophomore      F
## 157 FirstYear      F
## 158 Sophomore      F
## 159    Senior      F
## 160 Sophomore      F
## 161 FirstYear      M
## 162 FirstYear      F
## 163 Sophomore      M
## 164    Senior      M
## 165 Sophomore      M
## 166 Sophomore      F
## 167    Junior      M
## 168 FirstYear      M
```

```
## 169 Sophomore      M
## 170 Sophomore      M
## 171 Sophomore      F
## 172 Sophomore      F
## 173    Junior      F
## 174 Sophomore      F
## 175    Senior      F
## 176 FirstYear      M
## 177 Sophomore      M
## 178 Sophomore      F
## 179 FirstYear      M
## 180 Sophomore      F
## 181 Sophomore      M
## 182 Sophomore      F
## 183 Sophomore      F
## 184 Sophomore      F
## 185 FirstYear      F
## 186 Sophomore      M
## 187 Sophomore      M
## 188 Sophomore      F
## 189 Sophomore      M
## 190 Sophomore      M
## 191    Junior      M
## 192    Junior      F
## 193 Sophomore      M
## 194 Sophomore      F
## 195 Sophomore      M
## 196 Sophomore      M
## 197 Sophomore      F
## 198 Sophomore      M
## 199 FirstYear      M
## 200 FirstYear      F
## 201 Sophomore      F
## 202 FirstYear      M
## 203    Junior      F
## 204 Sophomore      F
## 205 Sophomore      F
## 206 FirstYear      F
## 207 Sophomore      M
## 208 FirstYear      M
## 209 Sophomore      M
## 210 FirstYear      M
## 211 Sophomore      F
## 212    Senior      M
## 213    Senior      F
## 214 Sophomore      F
## 215 Sophomore      M
## 216 FirstYear      F
## 217 Sophomore      F
## 218 Sophomore      F
```

```
## 219 Sophomore      M
## 220    Junior      M
## 221 Sophomore      M
## 222 Sophomore      M
## 223 Sophomore      M
## 224 Sophomore      M
## 225    Senior      M
## 226 Sophomore      F
## 227 Sophomore      F
## 228 Sophomore      M
## 229 Sophomore      F
## 230 FirstYear      F
## 231 Sophomore      F
## 232 Sophomore      M
## 233 Sophomore      M
## 234 Sophomore      M
## 235    Junior      F
## 236 Sophomore      F
## 237 Sophomore      M
## 238 FirstYear      M
## 239 Sophomore      M
## 240 Sophomore      M
## 241 Sophomore      F
## 242 Sophomore      M
## 243 FirstYear      M
## 244 Sophomore      F
## 245    Senior      M
## 246 Sophomore      F
## 247 FirstYear      M
## 248 Sophomore      M
## 249 Sophomore      M
## 250 Sophomore      M
## 251    Junior      M
## 252 FirstYear      M
## 253 Sophomore      M
## 254    Senior      M
## 255 Sophomore      M
## 256 Sophomore      F
## 257    Junior      F
## 258 Sophomore      M
## 259 Sophomore      M
## 260 Sophomore      M
## 261 Sophomore      M
## 262 FirstYear      M
## 263      <NA>      F
## 264    Junior      M
## 265 FirstYear      F
## 266    Junior      F
## 267 Sophomore      M
## 268    Junior      M
```

```
## 269 Sophomore      M
## 270 Sophomore      F
## 271 FirstYear      M
## 272 FirstYear      F
## 273 Sophomore      F
## 274 Sophomore      F
## 275 FirstYear      F
## 276 Sophomore      F
## 277    Senior      M
## 278    Senior      M
## 279 FirstYear      F
## 280 FirstYear      M
## 281 Sophomore      M
## 282 FirstYear      F
## 283    Senior      M
## 284 Sophomore      F
## 285 Sophomore      F
## 286 Sophomore      F
## 287    Senior      M
## 288 Sophomore      F
## 289 FirstYear      M
## 290 FirstYear      M
## 291 Sophomore      F
## 292 Sophomore      M
## 293    Junior      F
## 294 FirstYear      F
## 295 Sophomore      M
## 296 Sophomore      F
## 297    Senior      M
## 298 Sophomore      F
## 299 FirstYear      M
## 300 Sophomore      M
## 301 FirstYear      F
## 302 Sophomore      M
## 303 Sophomore      M
## 304 Sophomore      M
## 305 Sophomore      F
## 306 Sophomore      M
## 307    Junior      M
## 308    Junior      F
## 309 Sophomore      M
## 310 Sophomore      F
## 311    Senior      M
## 312 Sophomore      M
## 313 FirstYear      F
## 314    Junior      F
## 315 Sophomore      F
## 316 FirstYear      F
## 317    Senior      M
## 318 Sophomore      F
```

```
## 319 Sophomore      M
## 320    Senior      F
## 321    Junior      M
## 322    Senior      M
## 323 FirstYear      M
## 324    Senior      F
## 325    Junior      F
## 326 FirstYear      M
## 327 Sophomore      F
## 328 Sophomore      M
## 329 FirstYear      M
## 330 Sophomore      M
## 331    Junior      M
## 332 FirstYear      M
## 333    Junior      F
## 334 FirstYear      M
## 335    Junior      M
## 336 Sophomore      M
## 337 Sophomore      F
## 338 Sophomore      M
## 339 Sophomore      M
## 340 Sophomore      M
## 341 Sophomore      F
## 342 Sophomore      M
## 343    Senior      M
## 344 Sophomore      M
## 345 FirstYear      M
## 346 Sophomore      F
## 347 Sophomore      M
## 348 FirstYear      M
## 349 FirstYear      F
## 350    Senior      M
## 351 Sophomore      F
## 352 FirstYear      M
## 353 Sophomore      F
## 354 Sophomore      M
## 355    Senior      F
## 356    Junior      F
## 357 Sophomore      M
## 358 Sophomore      F
## 359 Sophomore      M
## 360 Sophomore      F
## 361 FirstYear      M
## 362 Sophomore      F
```

c) Convert the **Year** column to factors and properly order the factors based on common US progression (FirstYear - Sophomore - Junior - Senior)

```r
COI <- COI %>%
  mutate(Year = factor(Year,levels = c("FirstYear","Sophomore","Junior","Senior"), ordered = TRUE
```

d) Convert the **Gender** column to factors and rename them Male/Female.

```r
COI <- COI %>%
  mutate(Gender = factor(Gender)) %>%
  mutate(Gender = ifelse(Gender == "F","Female","Male"))
COI
```

```
##          Year Gender
## 1      Senior   Male
## 2   Sophomore Female
## 3   FirstYear   Male
## 4      Junior   Male
## 5   Sophomore Female
## 6   Sophomore Female
## 7   FirstYear Female
## 8   Sophomore   Male
## 9      Junior Female
## 10  FirstYear Female
## 11  Sophomore Female
## 12  FirstYear   Male
## 13  Sophomore   Male
## 14     Junior Female
## 15  FirstYear   Male
## 16  FirstYear Female
## 17  FirstYear Female
## 18  Sophomore   Male
## 19  Sophomore Female
## 20  Sophomore Female
## 21  Sophomore Female
## 22  Sophomore Female
## 23  FirstYear   Male
## 24  Sophomore   Male
## 25  Sophomore   Male
## 26  Sophomore   Male
## 27  Sophomore Female
## 28     Junior   Male
## 29  FirstYear   Male
## 30     Senior   Male
## 31     Junior   Male
## 32  Sophomore Female
## 33  FirstYear   Male
## 34     Senior   Male
## 35  FirstYear Female
## 36  Sophomore Female
## 37  FirstYear   Male
## 38  Sophomore Female
```

```
## 39  Sophomore Female
## 40     Junior Female
## 41     Senior   Male
## 42  Sophomore Female
## 43  Sophomore Female
## 44  FirstYear Female
## 45  FirstYear   Male
## 46     Junior   Male
## 47     Senior   Male
## 48  Sophomore   Male
## 49     Senior   Male
## 50  Sophomore Female
## 51  FirstYear   Male
## 52     Senior Female
## 53     Junior   Male
## 54     Senior   Male
## 55     Junior   Male
## 56  FirstYear   Male
## 57     Junior Female
## 58  Sophomore Female
## 59  Sophomore Female
## 60  FirstYear Female
## 61  Sophomore   Male
## 62     Senior   Male
## 63  FirstYear Female
## 64  FirstYear   Male
## 65  FirstYear   Male
## 66  Sophomore Female
## 67  FirstYear   Male
## 68  Sophomore   Male
## 69  Sophomore   Male
## 70     Senior   Male
## 71     Senior Female
## 72  Sophomore Female
## 73  FirstYear Female
## 74  FirstYear Female
## 75  Sophomore Female
## 76  FirstYear Female
## 77  FirstYear Female
## 78  Sophomore Female
## 79  Sophomore Female
## 80  Sophomore   Male
## 81  Sophomore   Male
## 82  FirstYear Female
## 83  FirstYear Female
## 84     Junior Female
## 85  FirstYear Female
## 86  Sophomore Female
## 87     Senior   Male
## 88  FirstYear Female
```

```
## 89  FirstYear   Male
## 90  FirstYear   Male
## 91  Sophomore   Male
## 92  FirstYear Female
## 93  Sophomore Female
## 94  Sophomore   Male
## 95  FirstYear   Male
## 96  Sophomore   Male
## 97  Sophomore Female
## 98  Sophomore   Male
## 99  Sophomore   Male
## 100 FirstYear   Male
## 101    Senior Female
## 102 Sophomore Female
## 103 FirstYear   Male
## 104 Sophomore   Male
## 105 Sophomore   Male
## 106      <NA> Female
## 107 Sophomore   Male
## 108 Sophomore   Male
## 109 Sophomore Female
## 110 Sophomore   Male
## 111 Sophomore   Male
## 112 Sophomore   Male
## 113 Sophomore Female
## 114 FirstYear   Male
## 115 FirstYear Female
## 116 Sophomore Female
## 117 Sophomore Female
## 118 Sophomore   Male
## 119 Sophomore   Male
## 120 FirstYear Female
## 121 Sophomore Female
## 122 FirstYear   Male
## 123 Sophomore   Male
## 124 Sophomore   Male
## 125 Sophomore Female
## 126 Sophomore   Male
## 127 Sophomore   Male
## 128 FirstYear   Male
## 129 Sophomore Female
## 130 Sophomore   Male
## 131    Senior Female
## 132 FirstYear Female
## 133 Sophomore Female
## 134 Sophomore Female
## 135 Sophomore Female
## 136 Sophomore Female
## 137 Sophomore   Male
## 138 FirstYear Female
```

```
## 139 Sophomore Female
## 140    Junior Female
## 141 FirstYear   Male
## 142 FirstYear Female
## 143 FirstYear   Male
## 144    Junior   Male
## 145 FirstYear Female
## 146 Sophomore Female
## 147 FirstYear Female
## 148 Sophomore   Male
## 149 Sophomore   Male
## 150 FirstYear Female
## 151 Sophomore Female
## 152 FirstYear   Male
## 153 FirstYear Female
## 154 Sophomore   Male
## 155    Senior   Male
## 156 Sophomore Female
## 157 FirstYear Female
## 158 Sophomore Female
## 159    Senior Female
## 160 Sophomore Female
## 161 FirstYear   Male
## 162 FirstYear Female
## 163 Sophomore   Male
## 164    Senior   Male
## 165 Sophomore   Male
## 166 Sophomore Female
## 167    Junior   Male
## 168 FirstYear   Male
## 169 Sophomore   Male
## 170 Sophomore   Male
## 171 Sophomore Female
## 172 Sophomore Female
## 173    Junior Female
## 174 Sophomore Female
## 175    Senior Female
## 176 FirstYear   Male
## 177 Sophomore   Male
## 178 Sophomore Female
## 179 FirstYear   Male
## 180 Sophomore Female
## 181 Sophomore   Male
## 182 Sophomore Female
## 183 Sophomore Female
## 184 Sophomore Female
## 185 FirstYear Female
## 186 Sophomore   Male
## 187 Sophomore   Male
## 188 Sophomore Female
```

```
## 189 Sophomore   Male
## 190 Sophomore   Male
## 191    Junior   Male
## 192    Junior Female
## 193 Sophomore   Male
## 194 Sophomore Female
## 195 Sophomore   Male
## 196 Sophomore   Male
## 197 Sophomore Female
## 198 Sophomore   Male
## 199 FirstYear   Male
## 200 FirstYear Female
## 201 Sophomore Female
## 202 FirstYear   Male
## 203    Junior Female
## 204 Sophomore Female
## 205 Sophomore Female
## 206 FirstYear Female
## 207 Sophomore   Male
## 208 FirstYear   Male
## 209 Sophomore   Male
## 210 FirstYear   Male
## 211 Sophomore Female
## 212    Senior   Male
## 213    Senior Female
## 214 Sophomore Female
## 215 Sophomore   Male
## 216 FirstYear Female
## 217 Sophomore Female
## 218 Sophomore Female
## 219 Sophomore   Male
## 220    Junior   Male
## 221 Sophomore   Male
## 222 Sophomore   Male
## 223 Sophomore   Male
## 224 Sophomore   Male
## 225    Senior   Male
## 226 Sophomore Female
## 227 Sophomore Female
## 228 Sophomore   Male
## 229 Sophomore Female
## 230 FirstYear Female
## 231 Sophomore Female
## 232 Sophomore   Male
## 233 Sophomore   Male
## 234 Sophomore   Male
## 235    Junior Female
## 236 Sophomore Female
## 237 Sophomore   Male
## 238 FirstYear   Male
```

```
## 239 Sophomore   Male
## 240 Sophomore   Male
## 241 Sophomore Female
## 242 Sophomore   Male
## 243 FirstYear   Male
## 244 Sophomore Female
## 245    Senior   Male
## 246 Sophomore Female
## 247 FirstYear   Male
## 248 Sophomore   Male
## 249 Sophomore   Male
## 250 Sophomore   Male
## 251    Junior   Male
## 252 FirstYear   Male
## 253 Sophomore   Male
## 254    Senior   Male
## 255 Sophomore   Male
## 256 Sophomore Female
## 257    Junior Female
## 258 Sophomore   Male
## 259 Sophomore   Male
## 260 Sophomore   Male
## 261 Sophomore   Male
## 262 FirstYear   Male
## 263      <NA> Female
## 264    Junior   Male
## 265 FirstYear Female
## 266    Junior Female
## 267 Sophomore   Male
## 268    Junior   Male
## 269 Sophomore   Male
## 270 Sophomore Female
## 271 FirstYear   Male
## 272 FirstYear Female
## 273 Sophomore Female
## 274 Sophomore Female
## 275 FirstYear Female
## 276 Sophomore Female
## 277    Senior   Male
## 278    Senior   Male
## 279 FirstYear Female
## 280 FirstYear   Male
## 281 Sophomore   Male
## 282 FirstYear Female
## 283    Senior   Male
## 284 Sophomore Female
## 285 Sophomore Female
## 286 Sophomore Female
## 287    Senior   Male
## 288 Sophomore Female
```

```
## 289 FirstYear   Male
## 290 FirstYear   Male
## 291 Sophomore Female
## 292 Sophomore   Male
## 293    Junior Female
## 294 FirstYear Female
## 295 Sophomore   Male
## 296 Sophomore Female
## 297    Senior   Male
## 298 Sophomore Female
## 299 FirstYear   Male
## 300 Sophomore   Male
## 301 FirstYear Female
## 302 Sophomore   Male
## 303 Sophomore   Male
## 304 Sophomore   Male
## 305 Sophomore Female
## 306 Sophomore   Male
## 307    Junior   Male
## 308    Junior Female
## 309 Sophomore   Male
## 310 Sophomore Female
## 311    Senior   Male
## 312 Sophomore   Male
## 313 FirstYear Female
## 314    Junior Female
## 315 Sophomore Female
## 316 FirstYear Female
## 317    Senior   Male
## 318 Sophomore Female
## 319 Sophomore   Male
## 320    Senior Female
## 321    Junior   Male
## 322    Senior   Male
## 323 FirstYear   Male
## 324    Senior Female
## 325    Junior Female
## 326 FirstYear   Male
## 327 Sophomore Female
## 328 Sophomore   Male
## 329 FirstYear   Male
## 330 Sophomore   Male
## 331    Junior   Male
## 332 FirstYear   Male
## 333    Junior Female
## 334 FirstYear   Male
## 335    Junior   Male
## 336 Sophomore   Male
## 337 Sophomore Female
## 338 Sophomore   Male
```

```
## 339 Sophomore   Male
## 340 Sophomore   Male
## 341 Sophomore Female
## 342 Sophomore   Male
## 343    Senior   Male
## 344 Sophomore   Male
## 345 FirstYear   Male
## 346 Sophomore Female
## 347 Sophomore   Male
## 348 FirstYear   Male
## 349 FirstYear Female
## 350    Senior   Male
## 351 Sophomore Female
## 352 FirstYear   Male
## 353 Sophomore Female
## 354 Sophomore   Male
## 355    Senior Female
## 356    Junior Female
## 357 Sophomore   Male
## 358 Sophomore Female
## 359 Sophomore   Male
## 360 Sophomore Female
## 361 FirstYear   Male
## 362 Sophomore Female
```

e) Produce a data set with eight rows and three columns that contains the number of responses for each gender:year combination. *You might want to look at the following functions:* ***`dplyr::count`*** *and* ***`dplyr::drop_na`****.*

```r
table <- COI %>%
  drop_na(Gender,Year) %>%
  count(Year,Gender) %>%
  rename(Responses = n)
table
```

```
##        Year Gender Responses
## 1 FirstYear Female        43
## 2 FirstYear   Male        51
## 3 Sophomore Female        96
## 4 Sophomore   Male        99
## 5    Junior Female        18
## 6    Junior   Male        17
## 7    Senior Female        10
## 8    Senior   Male        26
```

f) Pivot the table in part (e) to produce a table of the number of responses in the following form:

Gender	First Year	Sophomore	Junior	Senior
Female				
Male				

```
table3 <- table %>%
  pivot_wider(
    names_from = Year,
    values_from = Responses
  )
table3
```

```
## # A tibble: 2 x 5
##   Gender FirstYear Sophomore Junior Senior
##   <chr>      <int>     <int>  <int>  <int>
## 1 Female        43        96     18     10
## 2 Male          51        99     17     26
```

Problem 2

For this problem we will consider two simple data sets. Remember to erase eval = FALSE from the code chunk below.

```
    A <- tribble(
      ~Name, ~Car,
      'Alice', 'Ford F150',
      'Bob',   'Tesla Model III',
      'Charlie', 'VW Bug')

    B <- tribble(
      ~First.Name, ~Pet,
      'Bob',  'Cat',
      'Charlie', 'Dog',
      'Alice', 'Rabbit')
    A
```

```
## # A tibble: 3 x 2
##   Name    Car
##   <chr>   <chr>
## 1 Alice   Ford F150
## 2 Bob     Tesla Model III
## 3 Charlie VW Bug
```

```
    B
```

```
## # A tibble: 3 x 2
##   First.Name Pet
##   <chr>      <chr>
```

```
## 1 Bob        Cat
## 2 Charlie    Dog
## 3 Alice      Rabbit
```

a. Squish the data frames together to generate a data set with three rows and three columns. Do two ways: first using `cbind` and then using one of the `dplyr join` commands.

```r
B <- arrange(B, First.Name, Pet)
B
```

```
## # A tibble: 3 x 2
##   First.Name Pet
##   <chr>      <chr>
## 1 Alice      Rabbit
## 2 Bob        Cat
## 3 Charlie    Dog
```

```r
AB <- cbind(A,B[2])
AB
```

```
##      Name             Car    Pet
## 1   Alice       Ford F150 Rabbit
## 2     Bob Tesla Model III    Cat
## 3 Charlie          VW Bug    Dog
```

b. It turns out that Alice also has a pet guinea pig. Add another row to the `B` data set. Do this using either the base function `rbind`, or either of the `dplyr` functions `add_row` or `bind_rows`.

```r
pig <- c("Alice","Guinea Pig")
B <- rbind(B,pig)
```

c. Squish the `A` and `B` data sets together to generate a data set with four rows and three columns. Do this two ways: first using `cbind` and then using one of the `dplyr join` commands. Which was easier to program? Which is more likely to have an error.

```r
blank <- c("Alice","Ford F150")
A <- rbind(A,blank)
cbind(A,B[2])
```

```
##      Name             Car        Pet
## 1   Alice       Ford F150     Rabbit
## 2     Bob Tesla Model III        Cat
## 3 Charlie          VW Bug        Dog
## 4   Alice       Ford F150 Guinea Pig
```

```r
AB <- left_join(A,B, by = c("Name" = "First.Name"))
```

```
## Warning in left_join(A, B, by = c(Name = "First.Name")): Detected an unexpected many-to-many r
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 1 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

Using the dplyr function left_join was easier because I didn't have to code anything else. Cbind is more likely to have an error.

Problem 3

From this book's GitHub there is a .csv file of the daily maximum temperature in Flagstaff at the Pulliam Airport. The link is: https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-raw/FlagMaxTemp.csv

a) Create a line graph that gives the daily maximum temperature for 2005. *Make sure the x-axis is a date and covers the whole year.*

```r
MaxTemp <- read.csv("https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-raw/Flag
MaxTemp
```

```
##       X Year Month    X1    X2    X3    X4    X5    X6    X7    X8    X9   X10
## 1     1 1985     5 71.06 71.06 68.00 68.00 64.94 64.04 64.04 64.94 69.08 66.02
## 2     2 1985     6 62.96 62.96 64.94 60.08 69.08 75.92 82.04 86.00 84.92 84.02
## 3     3 1985     7 80.96 86.00 89.96 87.98 91.94 91.94 89.06 87.98 89.96 87.08
## 4     4 1985     8 77.00 68.00 78.08 80.06 82.04 80.96 82.94 82.94 80.06 80.06
## 5     5 1985     9 82.94 75.02 73.94 71.96 66.92 62.96 62.96 64.04 68.00 64.94
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## 7     7 1985    11 50.00 53.96 64.94 62.96 64.94 57.92 60.08 62.96 53.06 48.02
## 8     8 1985    12 33.08 42.98 44.06 46.04 50.00 51.08 46.94 51.08 39.92 35.96
## 9     9 1986     1 42.98 46.94 46.94 48.02 42.08 44.06 44.06 37.04 46.04 51.98
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## 11   11 1986     3 62.06 64.04 59.00 59.00 59.00 64.04 62.06 57.92 55.94 44.96
## 12   12 1986     4 62.06 46.04 35.96 44.96 53.06 57.02 55.04 57.02 55.04 60.08
## 13   13 1986     5 66.02 69.98 73.04 71.06 60.98 60.98 53.06 42.98 51.08 57.02
## 14   14 1986     6 75.02 78.98 71.06 73.94 75.02 73.94 73.04 73.04 68.00 69.08
## 15   15 1986     7 75.92 82.04 84.92 73.04 66.92 71.96 75.02 71.06 78.08 78.08
## 16   16 1986     8 87.08 86.00 86.00 84.02 82.94 82.04 80.06 82.04 78.98 80.06
## 17   17 1986     9 71.96 69.98 73.94 73.04 77.00 78.98 80.06 71.96 69.08 80.06
## 18   18 1986    10 60.08 60.08 53.06 50.00 57.02 51.98 57.92 64.94 62.96 62.96
## 19   19 1986    11 55.04 48.02 39.92 59.00 51.08 55.04 48.02 35.06 44.96 53.06
## 20   20 1986    12 48.02 50.00 57.02 59.00 53.96 50.00 35.06 39.92 41.00 35.96
## 21   21 1987     1 48.92 48.92 42.98 51.98 46.04 33.08 33.08 33.08 37.04 32.00
## 22   22 1987     2 39.02 55.94 55.94 46.94 39.92 42.08 42.98 46.94 51.98 51.08
## 23   23 1987     3 37.04 39.92 48.02 44.96 51.98 55.94 53.96 46.04 46.04 50.00
## 24   24 1987     4 55.94 60.08 59.00 53.06 42.08 50.00 53.06 50.00 53.96 60.98
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## 27   27 1987     7 80.06 78.08 82.04 82.04 80.06 80.06 80.96 84.02 80.96 78.08
## 28   28 1987     8 73.94 78.08 78.98 80.06 82.04 78.08 75.02 73.94 80.06 75.92
## 29   29 1987     9 82.04 78.98 78.08 75.02 62.06 68.00 73.04 78.98 75.92 77.00
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## 31   31 1987    12 46.94 50.00 53.96 64.94 55.04 39.02 51.98 46.94 42.98 55.04
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## 33   33 1988     2 41.00 35.96 37.94 35.96 32.00 39.92 50.00 48.02 51.98 59.00
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## 43   43 1988    12 41.00 44.06 55.04 48.92 44.96 46.94 50.00 39.02 33.08 37.94
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## 64   64 1990     9 80.96 64.04 71.96 73.04 71.96 69.08 73.04 75.92 78.08 80.96
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## 74   74 1991     7 82.04 87.08 87.08 87.98 87.98 82.04 82.04 82.04 77.00 75.02
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## 79   79 1991    12 26.96 26.06 39.02 37.04 44.06 39.92 57.02 48.02 48.92 46.94
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## 82   82 1992     3 53.06 48.92 44.06 44.06 44.06 46.04 48.02 48.92 33.98 39.02
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## 324 324 2012     7 84.02 84.92 80.96 71.96 78.98 78.98 86.00 82.94 82.94 87.08
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## 365 365 2015    12    NA 50.00 55.94    NA    NA    NA 62.96 62.06    NA    NA
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## 3   84.02 84.02 84.92 87.08 84.02 84.92 78.98 80.96 75.92 73.94 64.94 71.96
## 4   80.06 75.92 78.98 78.98 80.96 80.96 78.98 75.02 82.04 82.04 69.98 80.06
## 5   68.00 66.02 66.92 75.02 73.94 73.04 73.04 69.98 51.98 59.00 55.04 68.00
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## 7   46.94 41.00 26.06 26.96 26.96 39.92 51.08 37.04 37.04 26.96 48.02 46.04
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## 58  46.04 33.08 32.00 24.98 37.04 39.02 44.96 57.92 57.02 60.08 60.98 62.06
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## 61  69.08 71.06 73.04 71.06 69.08 68.00 71.96 77.00 80.96 80.96 84.02 87.08
## 62  78.08 82.04 82.94 75.02 73.94 75.92 73.94 75.02 80.06 84.02 82.94 78.98
## 63  80.96 78.08 75.02 64.04 60.98 64.04 66.92 71.96 73.04 71.96 71.06 73.94
## 64  82.94 84.02 87.98 84.02 82.94 75.02 69.98 69.08 57.02 64.04 60.08 66.02
## 65  68.00 66.92 68.00 66.02 69.98 69.98 64.04 68.00 66.02 55.94 48.02 48.92
## 66  59.00 57.92 60.98 66.02 64.04 62.06 62.06 60.08 57.02 53.06 39.92 42.98
## 67  57.92 51.08 46.04 42.08 42.98 42.08 39.02 32.00 42.08 28.94 21.92 12.92
## 68  39.92 39.02 50.00 51.08 41.00 44.96 32.00 35.96 42.08 42.98 30.02 26.06
## 69  51.98 48.92 53.06 55.04 57.02 57.92 48.02 46.04 39.92 44.06 55.04 57.02
## 70  46.94 35.96 44.96 44.96 32.00 28.94 33.98 46.94 51.08 33.08 37.94 30.02
## 71  57.92 39.92 42.98 46.94 55.04 55.04 55.94 55.04 55.94 59.00 60.98 55.94
## 72  48.92 51.08 64.04 66.02 55.94 62.06 66.92 64.04 62.06 66.02 66.02 62.96
## 73  75.92 66.92 69.98 69.98 71.06 73.94 77.00 80.96 80.96 75.92 73.94 73.94
## 74  77.00 77.00 80.06 82.04 80.06 80.06 80.06 82.04 82.04 75.92 71.96 71.06
## 75  75.02 64.04 73.94 80.06 80.96 78.98 82.04 80.06 75.92 78.98 82.04 84.92
## 76  66.92 68.00 69.08 71.96 71.96 75.02 75.02 80.06 80.06 73.94 71.06 71.96
## 77  75.92 75.92 71.96 77.00 77.00 78.08 75.92 75.92 75.02 75.02 69.98 66.02
## 78  51.98 50.00 50.00 59.00 44.96 33.98 35.06 41.00 41.00 37.94 55.94 53.06
## 79  44.96 37.94 44.06 48.02 42.08 46.94 48.92 48.02 37.04 35.06 37.94 33.98
## 80  39.92 44.06 30.02 35.06 35.96 30.02 46.94 48.02 33.08 30.92 32.00 41.00
## 81  37.94 39.02 39.02 35.96 39.92 41.00 33.98 39.02 44.06 48.92 53.96 55.04
## 82  42.98 48.92 57.02 55.04 55.04 53.96 51.08 46.04 44.96 51.08 53.06 39.92
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## 84  60.98 66.02 68.00 71.06 73.04 71.06 71.96 73.94 75.02 59.00 62.96 55.04
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## 88  78.98 77.00 75.92 71.96 73.94 73.94 75.92 78.08 77.00 73.04 73.94 77.00
## 89  73.04 73.94 75.92 73.04 64.94 66.02 66.02 71.96 71.06 69.08 69.08 62.96
## 90  44.06 39.92 48.02 51.08 55.04 60.98 55.04 53.96 51.98 48.92 33.98 44.06
## 91  55.04 48.02 28.04 28.04 33.98 30.92 28.04 35.06 26.06 24.08 44.06 46.04
## 92  35.06 33.98 42.08 39.92 37.94 37.04 42.08 42.08 39.02 35.06 44.96 50.00
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## 94  55.94 50.00 42.98 53.96 46.04 55.04 55.94 55.04 53.96 57.02 55.04 59.00
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## 98  80.06 82.04 75.92 75.02 77.00 78.08 75.02 75.92 75.92 78.98 77.00 78.08
## 99  77.00 80.06 80.06 78.08 75.02 75.92 78.08 80.06 80.96 80.06 77.00 73.94
## 100 80.06 80.06 69.98 64.04 66.02 68.00 66.92 64.94 62.96 69.98 71.06 73.94
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## 105 46.94 35.06 28.94 35.96 50.00 53.96 55.04 51.98 37.94 30.02 39.92 37.94
## 106 51.98 46.94 48.02 50.00 57.02 66.02 60.98 60.98 53.06 48.92 48.92 53.96
## 107 39.92 53.96 60.98 62.96 64.04 66.92 71.06 71.06 71.96 73.04 73.04 71.96
## 108 60.08 57.02 66.92 66.02 69.08 69.98 62.96 60.08 60.08 57.02 60.98 68.00
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## 110 87.08 86.00 84.92 87.08 84.92 84.02 87.08 87.08 78.08 69.08 78.98 82.94
## 111 84.02 82.94 78.98 80.96 91.94 78.08 80.06 82.04 80.06 75.92 78.98 78.08
## 112 73.04 71.96 68.00 66.92 66.92 69.08 71.96 73.04 75.02 75.02 77.00 66.92
## 113 69.98 64.94 66.02 64.04 55.04 37.04 42.08 48.92 53.06 57.92 60.08 64.94
## 114 53.96 51.98 37.94 39.92 37.04 50.00 42.98 35.96 28.04 44.06 48.92 46.04
## 115 46.04 44.96 42.08 33.98 39.02 42.08 48.92 48.02 48.92 55.94 53.06 50.00
## 116 46.04 37.04 35.96 48.92 55.04 39.02 30.02 30.92 37.04 35.96 42.98 30.02
## 117 44.96 42.98 42.08 41.00 39.02 42.98 46.04 48.02 51.98 51.98 55.04 57.92
## 118 57.92 42.98 44.06 55.04 57.02 60.98 64.04 55.94 60.08 60.98 60.08 53.96
## 119 48.02 64.94 64.94 62.06 55.94 44.06 46.04 33.98 33.98 37.04 35.06 39.02
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## 122 84.02 82.94 80.06 73.94 78.08 80.96 73.94 78.08 69.08 80.06 78.98 80.96
## 123 84.02 84.02 82.04 80.96 82.04 75.92 75.92 75.92 82.94 75.92 69.08 73.94
## 124 73.04 75.02 78.08 78.08 78.08 75.92 75.02 73.94 73.94 75.02 75.92 75.02
## 125 73.04 73.04 71.06 68.00 75.02 73.94 69.08 71.06 71.06 71.96 24.08 68.00
## 126 55.04 66.92 68.00 62.06 62.96 64.94 62.96 64.94 66.02 66.92 62.06 66.02
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## 133 84.02 80.96 78.98 73.04 73.94 80.96 80.06 82.94 82.94 84.02 82.04 80.06
## 134 75.02 78.98 80.96 77.00 78.98 73.94 75.02 78.08 80.96 84.92 87.08 87.08
## 135 84.92 87.98 87.08 87.98 84.92 84.02 86.00 78.98 78.98 82.04 78.08 78.98
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## 143 42.98    NA 44.96 48.92 57.92 62.96 64.04 66.02 66.02 64.94 69.08 69.98
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## 145 71.96 73.04 68.00 55.94 62.96 60.08 75.02 75.92 82.04 80.06 77.00 82.04
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## 153 39.92 41.00 42.98 42.08    NA 39.02 35.96 30.02 37.94 42.08    NA 39.92
## 154 46.04 55.94 55.94 48.92 42.08 51.08 53.96 48.02 39.02 53.06 53.96 60.98
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## 156 62.96 62.96 66.92 68.00 71.96 75.02 71.96 69.98 75.02 77.00    NA 77.00
## 157 84.92 86.00 87.08 86.00 87.08 91.94 93.92 89.96 87.98 87.98 82.04 78.98
## 158 78.08 80.96 82.94 80.96 78.98 80.06 77.00 75.02 75.02 77.00 80.06 78.08
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## 189 33.08 37.04 42.98 53.06 53.06 53.06 50.00 48.92 55.94 60.08 64.94 62.96
## 190 32.00 39.02    NA 55.94 57.92 62.96 71.06 71.06 69.98 62.06 51.08 37.94
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## 193 75.92 77.00 80.96 78.08 80.06 78.08 78.08 75.92 80.06 78.98 80.96 82.04
## 194 73.04 78.08 77.00 71.96 78.08 80.06 82.04 84.02 84.92 82.94 75.02 75.02
## 195 78.98 78.98 80.96 75.02 77.00 78.08 75.02 73.94 78.08 78.08 78.98 80.96
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## 197 62.06 60.08 59.00 51.98 55.04 57.92 62.06 59.00 57.92 55.04 57.92 60.08
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## 242 71.06 69.08 75.02 71.96 73.04 73.04 75.02 75.92 75.02 75.92 78.08 80.06
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## 252 78.08 78.98 78.08 82.04 82.04 73.94 78.08 84.02 87.08 84.02 82.94 84.02
## 253 80.06 82.94 84.02 87.08 89.96 87.08 84.92 86.00 84.02 82.04 82.94 87.08
## 254 77.00 73.94 71.96 73.04 75.02 71.06 73.94 75.02 78.08 80.06 78.98 80.96
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## 258 48.92 39.02 48.92    NA 55.94 60.08 59.00 46.94 32.00 28.04    NA    NA
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## 259 48.02 28.94 26.96 17.06 42.98 19.94 23.00 35.06 32.00 30.92 33.08 26.06
## 260 53.06 44.06 44.96 37.94 35.96 42.98 48.92 50.00 55.04 37.94 37.94 51.98
## 261 60.98 60.98 66.92 69.98 69.08 71.06 71.96 64.94 73.04 62.06 60.08 48.92
## 262 60.08    NA 57.02    NA 51.08 53.06 44.96 64.04 57.02 55.04 48.92 53.06
## 263 77.00 77.00 80.06 78.08 73.04 75.02 73.94 77.00    NA 75.92 69.98 66.02
## 264 78.08 71.06 75.02 82.04 84.92    NA 89.06 80.06 82.04 82.94 84.92 86.00
## 265 86.00 80.96 84.92    NA    NA 89.06 87.08 82.94 84.02 80.06 80.06 87.98
## 266 80.06 82.04 84.02 82.04 82.04 80.06 82.04 80.96 82.04 80.06 80.06 84.92
## 267 77.00 80.06 80.06 80.96 78.98 78.08 73.04 66.92 66.92 69.08 68.00 71.06
## 268 69.98 71.06 71.06 66.92 60.08 66.02 55.94 53.06 62.06 69.98 66.02 71.06
## 269 60.08 57.92 60.08 64.04 62.06 60.08 62.96 66.02 68.00 66.92 57.92    NA
## 270 33.08 30.02 35.96 39.02 26.06 33.08 41.00 44.06 42.98 37.94 44.06 30.02
## 271 41.00 42.98 39.92 35.06 35.06 44.06 17.96 28.94 30.02 37.04 42.98 37.04
## 272 57.02 51.98 48.92 51.08 33.08 42.98 46.94 50.00 53.06 50.00 41.00 37.94
## 273 53.06 57.02 57.92 55.94 51.08 42.98 32.00 44.06 50.00 55.04 55.04 57.92
## 274 48.02 57.92 55.94 66.02 69.98 64.94 53.06 60.08 66.02 62.06 57.02 59.00
## 275 69.98 69.98 64.94 48.92 60.98 62.96 69.08 73.94 80.06 84.02 82.94 62.96
## 276 80.06 75.02 75.92 80.96 84.02 84.02 84.92 84.02 84.02 86.00 87.08 87.08
## 277 84.02 71.96 77.00 73.94 71.06 71.06 73.94 80.96 82.94 84.92 71.96 75.02
## 278 71.96 80.06 80.96 82.04 80.06 80.96 75.02 75.02 78.08 77.00 77.00 80.06
## 279 64.94 69.98 73.04 73.94 75.02 75.92 71.06 68.00 69.08 73.94 71.96 71.06
## 280 62.06 48.92 48.02 48.02 57.92 64.04 69.08 69.98 68.00 68.00 66.92 68.00
## 281 37.04 46.04 59.00 69.08 62.96 57.02 60.98 62.96 66.02 64.94 62.96 62.96
## 282 55.04 60.08 48.92 37.94 28.94 30.02 30.92 33.98 33.98 33.98 37.94 41.00
## 283 37.04 48.92 48.92 53.06 46.94 48.02 44.96 44.06 44.06 44.06 51.98 46.04
## 284 28.94 37.94 33.08 35.96 35.06 44.96 41.00 37.04 39.02 37.94 55.04 51.98
## 285 53.06 50.00 46.04 46.94 50.00 51.08 64.04 62.96 64.04 62.96 59.00 53.06
## 286 41.00 55.04 60.08 55.94 46.94 37.94 48.02 60.98 64.94 69.98 73.04 71.96
## 287 64.94 66.92 66.92 66.92 69.08 66.02 66.02 71.06 73.94 66.92 73.04 78.08
## 288 84.92 87.08 86.00 89.96 84.92 86.00 91.94 87.98 87.98 80.06 77.00 82.94
## 289 84.92 69.98 69.98 78.08 78.98 80.96 82.04 84.02 84.92 89.06 89.06 75.02
## 290 75.92 78.08 73.94 71.96 71.96 69.98 69.98 71.06 71.06 73.94 78.08 64.04
## 291 62.06 60.98 57.92 64.94 69.98 71.06 75.92 73.04 66.92 55.04 62.06 62.96
## 292 62.96 53.06 42.98 42.98 39.02 46.04 51.08 55.04 55.04 57.02 50.00 55.94
## 293 39.02 35.96 35.06 39.02 37.04 48.02 44.96 44.06 39.02 50.00 48.02 41.00
## 294 41.00 53.96 46.94 42.98 44.06 48.02 44.96 33.08 35.96 35.96 26.96 26.96
## 295 39.02 46.04 48.92 42.98 46.04 41.00 48.02 48.92 44.06 37.04 33.98 28.94
## 296 48.92 53.96 48.02 44.06 44.06 53.06 55.04 55.04 48.02 44.06 48.02 53.96
## 297 60.08 51.08 51.08 55.94 64.94 62.06 66.02 68.00 62.06 59.00 48.02 37.94
## 298 48.02 53.06 55.94 60.98 66.92 69.98 64.94 62.96 68.00 71.96 71.06 62.06
## 299 64.94 64.94 60.08 71.96 78.08 77.00 77.00 77.00 75.92 77.00 75.92 77.00
## 300 82.04 84.92 84.02 84.92 89.96 87.08 84.02 86.00 84.02 82.94 80.06 78.08
## 301 78.08 80.06 80.96 80.96 84.92 77.00 78.98 75.92 78.08 80.96 82.94 78.08
## 302 75.92 75.02 78.98 78.08 78.08 78.98 80.96 82.04 82.04 78.08 75.92 66.92
## 303 68.00 68.00 69.98 71.06 69.98 68.00 64.94 62.96 62.96 59.00 51.98 46.94
## 304 42.98 42.08 51.98 53.06 51.08 53.06 53.96 59.00 55.94 51.98 37.94 35.06
## 305 57.02 57.02 64.94 64.04 55.94 41.00 35.96 35.96 35.96 39.92 41.00 41.00
## 306 35.06 39.02 51.98 48.92 48.92 53.96 55.94 53.96 51.98 48.02 48.92 50.00
## 307 44.96 53.96 60.98 55.04 55.04 50.00 44.96 42.98 35.96 28.04 30.02 37.94
## 308 59.00 57.92 59.00 59.00 62.06 66.02 62.96 57.02 51.98 50.00 44.06 37.04
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## 309 53.06 57.92 55.94 53.06 59.00 68.00 68.00 62.96 62.96    NA 62.96 62.96
## 310 55.94 68.00 71.06 66.92 64.04 60.98 57.92 53.06 46.94 57.92 66.02 64.94
## 311 73.04 73.94 75.02 80.96 82.04 77.00 77.00 75.92 73.04 78.08 84.02 87.98
## 312 75.92 75.02 75.02 77.00 82.04 87.08 80.96    NA 78.98 84.02 82.94 82.94
## 313 82.04 84.02 80.06 78.08 82.04 84.02 86.00 84.02 80.96 80.06 82.04 84.92
## 314 71.06 66.02 66.02 64.04 62.96 66.92 68.00 71.06    NA    NA    NA 77.00
## 315 68.00 68.00 71.96 75.02 73.94 73.04 71.06 69.98 68.00 66.02 66.02 69.98
## 316 48.02 44.96 44.96 50.00 51.98 51.08 55.94 51.08 46.94 46.94 44.96 53.96
## 317 39.02 39.02 35.06 35.96 33.98 30.92 37.04 37.04 41.00 37.94 41.00 39.02
## 318 51.08 37.04 44.96 50.00 42.98 37.04 41.00 48.92 57.92 39.92 37.94 37.94
## 319 50.00 50.00 33.08 35.96 30.02 37.94 48.02 46.94 44.96 44.06 51.98 55.04
## 320 57.02 55.94 57.92 59.00 62.06 60.08 51.98 50.00 28.04 39.92 55.04 62.06
## 321 53.06 50.00 48.92 39.92 48.02 59.00 62.96 66.02 69.98 71.06 75.02 78.08
## 322 71.06 73.94 75.92 73.04 75.02 75.92 68.00 71.96 78.08 80.06 80.96 75.92
## 323 78.08 80.96 80.96 80.96 78.08 77.00 80.96 84.92 82.94 84.02 84.92 84.02
## 324 82.94 80.06 78.08 68.00 73.94 75.02 75.92 80.96 84.92 84.02 80.96 82.04
## 325 84.92 82.94 84.02 84.92 82.04 80.96 75.02 78.98 80.06 78.08 73.04 71.96
## 326 71.06 68.00 68.00 71.06 73.04 73.94 75.02 77.00 75.92 77.00 78.08 78.08
## 327    NA    NA    NA    NA 69.98 73.04 71.06 71.06 75.02 64.94 60.98 59.00
## 328 41.00 50.00 55.94 51.98 55.94 53.06 51.08 51.08 55.04 57.92 57.92 60.98
## 329 48.92 44.06 42.98 37.04 28.94 30.02 24.08 33.08 32.00 37.04 46.94 44.96
## 330 35.06 21.92 23.00 26.96 37.04 37.04 35.96 42.08 42.08 44.96 51.98 53.06
## 331 33.08 39.92 41.00 46.94 51.08 51.08 48.02 46.94 42.08 30.92 35.06 41.00
## 332 53.96 59.00 64.94 66.02 64.04 60.98 60.08 60.08 55.94 59.00 59.00 55.04
## 333 59.00 62.06 62.06 60.08 55.94 50.00 46.94 59.00 62.06 66.02 68.00 68.00
## 334 73.04 77.00 75.92 71.06 69.98 69.98 64.04 69.08 68.00 69.98 71.06 71.06
## 335 84.02 84.02 82.94 80.96 80.06 80.06 80.06 80.06 78.98 78.98 78.08 78.08
## 336 75.02 78.98 80.96 80.06 75.02 73.94 75.92 80.06 78.08 73.04 71.96 75.92
## 337 73.04 78.08 80.06 80.96 77.00 80.96 78.08 80.06 78.08 80.96 69.98 75.92
## 338 64.94 71.06 73.04 73.04 73.04 73.04 69.98 71.96 73.04 73.94 73.04 60.08
## 339 57.02 57.02 57.02 57.02 55.04 60.08 62.06 60.98 62.96 64.94    NA 62.06
## 340 57.92 59.00 60.08 55.04 53.06 42.08 44.06 55.04 51.08 46.94 41.00 39.92
## 341 30.02 35.96 44.96 37.04 48.02 48.92 44.06 53.96 51.08 37.04    NA 41.00
## 342 53.96 42.98 48.92 50.00 55.94 55.04 50.00 51.98 55.04 55.04 41.00 53.06
## 343 51.08 59.00 64.04 62.96 62.06 55.94 57.92 57.02 37.94 42.08    NA    NA
## 344    NA    NA    NA    NA    NA    NA    NA    NA    NA    NA    NA    NA
## 345 68.00 64.04 60.08 55.04 62.96 66.02 66.92 59.00 59.00 64.04 69.98 66.02
## 346 51.08 57.02 57.92 60.98 73.94 77.00 77.00 75.02 69.08 69.08 64.04 64.04
## 347 75.02 78.98 78.08 75.92 73.94 75.02 73.04 73.04 77.00 82.04 80.96 80.06
## 348 80.06 84.02 82.04 82.04    NA 78.08 82.94 82.94 78.08 80.06 82.94 84.92
## 349 75.92 71.06 64.94 75.02 77.00 80.96 80.96 80.06    NA 71.06 71.96 66.92
## 350 75.02 77.00 77.00 77.00 75.92 77.00 77.00 73.94 69.98 78.08 71.06 73.04
## 351 68.00 66.02 62.06 71.96 71.06 66.92 64.04 66.02 64.94 62.96 64.94 66.92
## 352 55.94 53.96 53.96 46.94 48.02 39.92 42.98 48.02 55.04 53.96 48.92 53.06
## 353 53.06 48.92 53.06 37.94 42.08 44.96 44.96 33.98 39.02 26.96 51.08 50.00
## 354 44.06 42.08 42.98 39.02 42.98 57.02 55.04 59.00 57.92 51.08    NA    NA
## 355 48.92 51.98 59.00 64.94    NA    NA    NA 62.96 62.96 57.02 53.96 48.02
## 356 57.02    NA 55.04 57.92 62.06 64.04 66.92    NA    NA 57.92 62.06 60.98
## 357 60.98 64.94 68.00 66.92 62.06 46.94 48.02 62.06 60.98 64.04 62.96 60.98
## 358 66.02 64.04 60.08 59.00 51.98 48.02 57.02 57.02 60.08 62.96 62.06 59.00
```

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## 359 71.06 73.04 73.94 77.00 80.96 82.04 82.94 84.02 84.02 87.98    NA 84.92
## 360 71.96 77.00 78.98 77.00 77.00 80.06 78.98 75.92 73.94 73.04 75.92 77.00
## 361 71.96 73.94 80.96 84.92 84.92 86.00 84.02 84.02 84.92 87.08 80.06 80.06
## 362 82.04 80.96 80.06 73.94 69.98 68.00 71.96 75.92 78.08 80.06 75.92 75.02
## 363 73.04 77.00 77.00 77.00 75.92 73.94 64.04 60.98 57.92 55.94 55.04 51.08
## 364 44.96 46.04 48.92 53.06 51.08 39.02 46.04 51.98 62.96 60.98 57.92 51.08
## 365    NA    NA    NA    NA    NA    NA 42.08 53.06    NA    NA    NA    NA
##       X23   X24   X25   X26   X27   X28   X29   X30   X31
## 1      NA 69.98 73.94 71.06 71.06 69.08 73.04 69.08 62.06
## 2   84.92 80.96 73.94 71.96 73.04 80.96 84.02 82.04    NA
## 3   73.04 80.06 80.06 77.00 80.06 82.04 77.00 73.94 73.04
## 4   84.02 87.98 91.04 84.92 84.02 80.96 78.98 82.04 82.94
## 5   68.00 71.06 71.96 73.04 71.96 64.94 60.98 64.04    NA
## 6   53.96 68.00 66.92 64.94 66.92 68.00 64.04 64.94 62.06
## 7   46.04 48.92 50.00 39.92 46.04 41.00 42.98 35.96    NA
## 8   57.02 55.04 51.98 46.94 55.94 51.98 53.96 53.06 44.96
## 9   53.06 53.06 50.00 48.92 51.08 57.02 60.98 59.00 46.94
## 10  55.04 60.08 62.96 69.08 69.98 60.98    NA    NA    NA
## 11  59.00 60.98 57.92 60.98 64.94 69.08 64.94 53.96 64.94
## 12  66.92 59.00 59.00 59.00 55.94 62.96 66.92 68.00    NA
## 13  71.06 69.98 75.02 80.06 80.96 78.98 73.04 73.94 64.94
## 14  82.04 84.92 71.06 78.08 84.92 86.00 82.94 71.06    NA
## 15  66.92 71.06 73.94 75.92 75.92 77.00 80.96 82.94 84.02
## 16  78.98 77.00 64.94 71.96 73.04 73.94 69.08 71.96 73.94
## 17  57.92 51.08 39.02 44.96 55.04 57.02 57.02 57.92    NA
## 18  55.94 59.00 60.08 60.98 60.08 60.08 64.94 64.94 62.06
## 19  44.06 39.92 48.92 48.02 42.98 50.00 59.00 50.00    NA
## 20  39.02 35.06 41.00 37.04 39.02 42.08 37.04 41.00 51.98
## 21  46.94 46.94 48.02 53.96 57.92 55.04 48.02 48.92 46.94
## 22  42.08 33.08 24.98 24.98 28.04 33.08    NA    NA    NA
## 23  28.04 39.02 35.06 39.02 46.94 37.04 35.06 37.04 48.02
## 24  64.04 69.08 69.08 69.08 68.00 68.00 68.00 66.02    NA
## 25  64.04 64.04 62.96 53.06 57.92 60.08 64.04 71.06 68.00
## 26  78.98 80.96 84.02 78.98 75.02 78.98 84.02 80.06    NA
## 27  78.98 82.94 87.08 77.00 73.94 73.94 62.06 78.08 71.06
## 28  80.06 71.96 73.04 73.94 66.92 75.02 75.02 75.92 80.96
## 29  71.06 62.06 62.06 60.08 71.96 71.06 69.98 71.96    NA
## 30  51.08 48.02 46.94 39.02 39.02 39.02 46.04 39.92    NA
## 31  46.94 30.02 26.06 19.04 21.92 41.00 37.94 39.92 28.94
## 32  33.98 51.98 32.00 42.98 48.92 50.00 57.02 50.00 44.96
## 33  53.96 55.04 53.96 53.06 46.04 51.08 53.96    NA    NA
## 34  60.08 62.96 59.00 69.98 69.08 66.02 46.04 55.04 46.94
## 35  37.94 44.96 48.92 57.02 62.06 64.04 55.94 64.94    NA
## 36  75.02 75.02 75.02 75.92 75.92 73.04 71.96 55.04 48.02
## 37  86.00 84.92 89.06 75.02 73.94 77.00 60.08 78.08    NA
## 38  80.96 87.08 82.94 84.02 84.92 80.06 77.00 82.04 73.94
## 39  75.92 77.00 78.98 75.92 78.08 60.08 71.06 71.96 64.94
## 40  51.98 66.02 68.00 69.08 69.98 71.96 73.04 66.92    NA
## 41  71.06 73.04 71.96 71.06 69.98 66.02 62.06 60.98 64.94
## 42  51.98 53.06 37.94 26.06 30.02 35.96 55.04 42.08    NA
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## 43  35.06 28.04 35.06 30.02 23.00 26.96 23.00 21.92 44.06
## 44  48.92 44.06 39.92 33.98 35.06 33.98 39.92 37.94 46.04
## 45  55.04 60.98 62.06 60.08 60.08 51.08    NA    NA    NA
## 46  60.98 59.00 57.92 53.96 48.02 50.00 60.98 57.92 60.98
## 47  68.00 62.06 57.02 53.96 42.98 53.96 57.92 62.06    NA
## 48  78.08 78.98 78.08 69.98 73.94 73.04 69.98 66.02 62.06
## 49  78.08 77.00 75.92 75.92 78.98 78.98 78.08 80.06    NA
## 50  77.00 78.98 78.08 78.08 73.04 73.04 69.08 71.06 78.98
## 51  73.94 75.02 71.96 71.96 73.94 73.94 78.98 80.96 78.08
## 52  78.08 78.08 75.92 78.08 75.92 75.92 78.08 77.00    NA
## 53  57.92 60.08 64.94 55.94 53.96 55.94 53.06 42.98 48.92
## 54  59.00 59.00 50.00 48.92 46.94 39.92 32.00 35.06    NA
## 55  50.00 59.00 55.04 51.98 50.00 51.98 44.96 32.00 37.94
## 56  46.04 51.08 37.94 46.04 48.92 33.08 44.06 42.98 46.94
## 57  46.94 55.94 57.92 57.92 51.08 51.98    NA    NA    NA
## 58  62.96 64.04 64.04 64.04 57.92 55.94 50.00 37.94 42.98
## 59  64.04 55.04 44.96 51.98 64.94 68.00 68.00 60.08    NA
## 60  73.94 71.96 69.98 68.00 66.92 69.08 57.02 53.06 62.06
## 61  86.00 87.08 87.98 91.04 91.04 89.96 89.06 89.06    NA
## 62  78.08 78.08 71.96 71.96 73.94 78.08 80.06 82.94 89.06
## 63  73.94 75.02 71.96 73.04 75.02 80.96 80.06 80.96 80.06
## 64  64.04 66.92 68.00 69.08 69.08 68.00 64.04 64.04    NA
## 65  66.02 64.04 64.94 69.98 71.06 68.00 68.00 66.02 64.94
## 66  46.04 53.96 62.06 55.94 35.96 33.08 32.00 37.04    NA
## 67  15.98 12.02 21.92 35.96 35.06 35.96 35.96 28.94 37.94
## 68  35.96 42.98 41.00 48.02 46.04 44.96 41.00 26.96 44.96
## 69  57.92 57.92 53.96 55.94 51.98 51.08    NA    NA    NA
## 70  39.02 48.02 51.08 42.98 30.92 30.92 39.92 35.96 46.04
## 71  55.94 57.02 59.00 53.06 51.98 51.08 51.08 57.92    NA
## 72  60.98 66.92 71.96 73.04 69.08 66.02 66.92 68.00 62.96
## 73  75.02 75.02 71.06 73.04 71.06 71.96 71.96 73.94    NA
## 74  78.08 80.96 78.98 80.06 75.02 87.98 86.00 80.96 75.92
## 75  82.94 78.98 80.06 75.92 80.06 73.04 78.08 82.04 82.94
## 76  75.92 73.94 78.08 80.06 78.98 75.02 71.96 69.98    NA
## 77  57.92 55.94 55.04 53.06 55.04 39.02 37.04 42.08 35.06
## 78  41.00 32.00 50.00 51.08 57.92 53.06 37.94 30.92    NA
## 79  35.96 37.94 30.02 37.04 39.92 28.94 37.94 37.94 37.04
## 80  44.06 42.98 57.02 46.94 55.94 46.94 48.02 50.00 44.96
## 81  55.94 42.98 48.92 46.04 44.06 53.96 57.92    NA    NA
## 82  42.98 42.98 50.00 55.04 51.98 39.92 46.94 48.92 53.06
## 83  60.08 68.00 69.98 73.04 73.94 75.02 77.00 75.02    NA
## 84  57.92 51.98 57.92 62.96 62.06 64.94 64.94 57.92 62.96
## 85  78.98 69.98 78.08 77.00 77.00 84.92 82.04 75.92    NA
## 86  77.00 78.08 68.00 73.04 78.08 78.98 80.06 84.02 84.02
## 87  55.94 51.98 66.02 69.98 71.96 75.92 75.92 71.06 71.06
## 88  78.08 75.92 73.94 75.02 75.02 77.00 77.00 77.00    NA
## 89  66.92 66.02 57.02 57.92 62.96 60.98 51.08 46.94 46.04
## 90  44.06 35.06 35.96 30.92 42.08 48.02 44.06 39.92    NA
## 91  41.00 37.04 46.04 35.96 44.96 46.04 41.00 39.02 39.02
## 92  53.96 35.96 46.94 46.04 46.04 53.06 51.98 42.08 42.08
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## 93  42.08 46.04 35.06 37.94 44.06 44.96    NA    NA    NA
## 94  62.06 62.06 55.04 57.02 48.92 35.96 37.94 44.06 53.96
## 95  66.92 62.06 60.98 66.92 69.08 66.02 69.08 71.06    NA
## 96  69.08 69.98 71.96 75.02 73.94 73.94 69.08 71.96 78.08
## 97  78.08 80.96 82.94 84.92 89.06 84.92 84.02 82.04    NA
## 98  75.92 75.92 77.00 78.08 80.06 84.02 86.00 82.94 82.94
## 99  77.00 80.06 78.98 66.92 68.00 62.96 69.98 69.98 69.98
## 100 75.02 73.04 73.04 75.02 73.04 78.98 78.98 77.00    NA
## 101 64.04 66.02 66.92 64.94 57.02 50.00 59.00 57.92 53.06
## 102 48.02 44.96 42.98 35.06 39.92 46.94 57.92 51.08    NA
## 103 37.04 26.06 39.92 42.98 48.92 39.92 44.96 37.94 41.00
## 104 55.94 51.08 48.92 35.96 35.96 35.06 37.04 37.94 35.96
## 105 33.08 44.96 51.08 57.02 53.06 50.00    NA    NA    NA
## 106 51.08 48.92 48.92 42.08 37.94 42.98 55.04 60.08 55.04
## 107 68.00 62.06 42.98 37.94 41.00 42.98 46.94 51.98    NA
## 108 69.98 73.04 62.96 53.06 62.96 69.98 71.96 77.00 78.08
## 109 84.02 87.08 91.04 91.04 91.04 87.08 91.04 91.04    NA
## 110 80.06 80.06 73.04 80.96 86.00 86.00 82.04 82.94 82.04
## 111 77.00 80.06 80.96 82.04 82.04 82.94 80.96 80.96 80.06
## 112 71.96 75.92 75.92 77.00 77.00 80.06 78.08 75.02    NA
## 113 64.94 64.04 62.96 60.08 62.06 68.00 69.08 62.96 62.96
## 114 37.04 46.04 53.96 48.92 32.00 32.00 37.04 42.08    NA
## 115 44.06 37.04 39.02 41.00 46.04 51.08 46.94 42.08 33.98
## 116 33.08 42.98 46.04 35.06 35.96 35.96 42.98 37.04 42.08
## 117 51.98 55.94 51.08 50.00 51.98 51.98    NA    NA    NA
## 118 48.92 48.02 39.02 35.96 46.04 51.08 46.04 44.96 46.04
## 119 53.96 53.96 57.92 64.04 64.94 64.04 62.96 64.94    NA
## 120 69.98 60.08 57.92 57.92 60.08 62.06 60.98 60.08 68.00
## 121 75.92 80.06 82.94 84.02 82.94 78.98 80.06 78.08    NA
## 122 78.98 82.94 84.92 87.98 89.06 91.04 91.94 89.06 84.92
## 123 71.96 77.00 77.00 75.92 78.98 78.98 80.96 82.04 82.94
## 124 71.06 69.08 69.98 69.08 71.06 69.08 62.06 62.96    NA
## 125 51.08 46.04 59.00 64.94 69.08 75.92 71.96 64.94 64.04
## 126 60.98 66.02 64.04 66.02 55.94 41.00 48.92 62.06    NA
## 127 35.06 30.92 32.00 32.00 39.02 48.92 44.06 44.06 46.94
## 128 26.96 33.08 39.02 37.94 39.92 44.06 44.96 46.94 51.98
## 129 42.98 46.04 42.98 33.98 24.08 30.02 33.08    NA    NA
## 130 59.00 42.98 48.02 50.00 53.06 55.94 53.96 46.94 60.08
## 131 66.02 71.96 73.94 78.08 80.06 77.00 55.94 66.02    NA
## 132 71.06 64.94 62.06 48.92 57.02 66.02 71.06 71.06 69.98
## 133 77.00 78.98 78.08 77.00 71.96 69.98 78.08 84.02    NA
## 134 91.04 93.92 89.06 86.00 87.98 82.94 78.08 82.94 82.94
## 135 75.02 77.00 75.92 75.92 77.00 80.96 84.92 86.00 87.08
## 136 69.08 71.06 69.08 66.02 68.00 59.00 66.02 73.04    NA
## 137 48.92 55.94 55.04 41.00 30.02 33.98 37.04 44.06 50.00
## 138 41.00 44.96 48.92 51.08 33.98 37.94 48.02 32.00    NA
## 139 39.92 42.08 44.06 53.06 46.04 46.04 44.06 57.02 53.06
## 140 35.96 35.96 44.96    NA 44.96 46.04 46.94 39.02    NA
## 141 44.06 42.08 32.00 37.04 39.92 28.04    NA    NA    NA
## 142 66.92 59.00 59.00 50.00 55.94 57.92 57.92 55.04 60.08
```

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## 143 66.02 64.04 44.06 50.00 64.94 64.04 66.92 60.08    NA
## 144 69.08 66.02 64.04 64.04 69.08 73.94 78.98 82.04 82.94
## 145 78.08 77.00 78.98 82.04 80.06 78.08 75.92 75.92    NA
## 146 75.02 78.98 84.02 87.08 78.08 78.98 78.98 80.06 78.08
## 147 78.98 82.94 80.96 78.08 66.92 78.98 78.08 75.02    NA
## 148 69.98 69.98 71.96 64.94 66.02 73.94 73.04 73.94    NA
## 149 50.00 53.96 44.06 42.98 51.08 57.92 60.08 57.92 64.04
## 150 46.04 57.02 57.92 62.06 55.04 37.94 51.08 55.04    NA
## 151 28.04 37.04    NA 21.92 28.04 44.06 39.02 46.04 44.96
## 152 48.02 46.94 51.98    NA 53.06 53.06 55.94 48.02 33.98
## 153 44.06    NA 28.04 30.92 33.08 35.06    NA    NA    NA
## 154 62.96 66.02 64.04 57.02 37.04 39.02 37.94 28.94 33.98
## 155 69.08 66.02 57.02 51.98 53.06 55.94 62.06 55.04    NA
## 156 73.94 75.02 77.00 78.98 80.06 84.92 87.98 87.98    NA
## 157 75.92 71.96 68.00 78.08 82.94 82.04 84.02 82.04 80.06
## 158 80.06 82.04 80.06 73.94 78.98 84.02 87.08 84.02    NA
## 159 64.94 68.00 69.08 64.94 64.04 69.08 71.06 69.08    NA
## 160 60.08 59.00 59.00 46.94 46.04 51.08 57.02 53.06 44.06
## 161 59.00 57.02 51.98 62.06 64.04 62.06 39.92 48.92    NA
## 162 37.04 33.98 44.06 48.92 53.06 51.98 57.92 57.92 60.08
## 163 48.02 53.96 53.96 46.94 32.00 33.98 33.98 46.04 48.92
## 164 48.02 55.94 60.08 55.04 53.06 64.94    NA    NA    NA
## 165 62.06 59.00 60.98 57.92 60.08 60.08 60.98 64.04 60.08
## 166 53.96 44.06 42.08 48.92 66.02 64.94 57.02 39.92    NA
## 167 69.98 68.00 64.94 62.06 71.06 73.04 71.96 71.96 71.06
## 168 80.96 82.04 80.96 82.04 82.04 82.04 84.92 87.98    NA
## 169 77.00 80.96 77.00 73.04 80.06 59.00 69.98 71.06 69.08
## 170 82.04 80.06 75.92 77.00 73.04 75.02 71.96 73.04 71.96
## 171 64.04 57.92 66.02 68.00 73.04 71.06 57.92 66.02    NA
## 172 69.98 66.02 64.94 66.92 64.04 62.96 59.00 55.94 55.94
## 173 39.92    NA 41.00 53.96 64.04 53.96 57.02 57.02    NA
## 174 35.96 35.96 41.00 42.08 35.06 42.98 46.94 53.06 51.98
## 175 46.04 50.00 48.92 44.96 35.96 42.08 39.92 39.92 42.08
## 176 37.04 39.92 33.98 33.08 48.02 51.98 35.96    NA    NA
## 177 37.94 50.00 50.00 53.06 60.98 57.92 44.96    NA 51.08
## 178 55.04 64.04 66.92 71.06 75.92 73.94 69.08 62.06    NA
## 179 82.04 82.94 80.96 71.06 73.04 80.06 84.02 84.02 82.04
## 180 71.06 62.96 71.96 77.00 78.98 78.98 80.96 78.98    NA
## 181 86.00 86.00 87.08 87.98 86.00 86.00 89.96 91.04 91.94
## 182 80.06 78.98 78.08 75.02 75.92 73.04 66.92 64.94 68.00
## 183 69.98 62.96 62.06 73.04 66.92 75.02 77.00 73.94    NA
## 184 48.02 46.04 46.94 55.94 53.96 46.94 39.92 44.96 42.98
## 185 46.94 48.02 50.00 48.02 57.92 55.04 57.02 55.94    NA
## 186 53.06 53.96 50.00 33.08 37.94 50.00 51.08 53.06 55.94
## 187 48.92 46.04 44.96 39.02 37.94 30.92 28.94 35.06 28.94
## 188 50.00 35.06 37.04 42.98 37.94 41.00    NA    NA    NA
## 189 62.06 62.06 62.96 62.96 60.08 57.02 60.08 60.08 62.06
## 190 50.00 57.02 66.02 71.96 73.04    NA 66.02 66.92    NA
## 191 75.92 82.94 82.94 82.04 78.98 75.02 71.06 77.00 82.94
## 192 82.94 78.98 78.98 80.06 71.06 80.06 84.92 89.06    NA
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## 193 82.94 82.04 82.04 69.98 77.00 82.94 87.98 84.92 75.02
## 194 73.94 78.98 80.96 86.00 87.98 87.08 84.92 80.96 78.08
## 195 80.06 80.96 80.96 78.08 80.96 80.96 80.96 78.98    NA
## 196 64.94 66.02 69.98 73.04 71.96 69.98 64.94 66.92 68.00
## 197 51.08 44.06 46.04 35.96 37.04 28.04 30.02 30.02    NA
## 198 35.96 30.02 37.94 46.04 48.92 46.94 46.94 39.92 46.94
## 199 41.00 30.02 41.00 53.96 51.08 44.06 37.04 32.00 26.06
## 200 64.04 57.92 55.94 57.92 46.04 55.04    NA    NA    NA
## 201 62.96 50.00 41.00 48.92 60.08 62.96 62.06 62.06 66.02
## 202 69.98 69.08 73.94 68.00 57.02 51.98 66.02 66.92    NA
## 203 62.96 64.04 69.98 73.04 69.98 73.94 78.08 84.92 87.08
## 204 80.96 84.92 87.08 87.08 87.98 84.92 84.02 84.92    NA
## 205 87.08 82.04 78.08 82.94 82.94 80.96 82.94 89.96 86.00
## 206 77.00 78.98 82.94 84.02 84.92 84.02 80.06 75.02 80.06
## 207 78.98 78.98 78.08 75.92 73.94 64.04 64.04 62.06    NA
## 208 57.02 51.98 53.96 53.96 44.96 44.06 51.08 48.02 51.98
## 209    NA 64.94 55.04 39.92 39.02 42.98 44.96 53.06    NA
## 210 26.96 26.96 28.94 35.06 42.08 42.08 53.06 39.92 44.96
## 211 51.08 60.08 57.02 53.96 57.92 62.06 55.94 57.92 60.98
## 212 51.98 48.92 46.04 37.04 35.06 33.08    NA    NA    NA
## 213 59.00 60.08 55.94 57.92 62.06 48.92 39.92 48.02 59.00
## 214 39.02 50.00 59.00 62.96 62.96 64.94 60.98 57.92    NA
## 215 80.06 78.08 78.08 78.08 80.06 84.92 82.94 84.02 80.96
## 216 73.94 71.96 71.06 77.00 82.94 84.92 86.00 87.98    NA
## 217 87.08 84.02 87.08 82.94 82.04 84.92 73.04 80.06 82.04
## 218 82.04 82.04 75.92 78.98 80.06 66.02 73.04 75.92 80.06
## 219 78.98 78.08 71.06 78.98 82.94 82.94 82.04 78.98    NA
## 220 78.08 77.00 73.04 62.96 57.02 66.92 71.96 62.96 55.04
## 221 33.08 35.06 51.08 44.96    NA 42.98 39.92 64.04    NA
## 222 42.98 42.08 37.94 37.04 28.94 23.00 30.92 35.96 41.00
## 223 37.94 44.96    NA 39.02 35.96 42.98 39.92 48.92 46.94
## 224 33.98 37.04 37.94 46.04 44.96 37.04 28.94    NA    NA
## 225 69.98 64.94 64.94 64.04 55.94 57.92 55.94 60.98 66.92
## 226    NA 51.08 60.98 66.02 68.00 73.04 66.92 48.92    NA
## 227 64.04 64.94 64.94 64.94 66.02 73.94 71.96 66.92 71.96
## 228 77.00 75.92 78.08 82.04 82.04 80.96 77.00 73.94    NA
## 229 86.00 82.94 77.00 82.04 75.92 78.08 80.06 82.94 82.94
## 230 73.94 71.06 68.00 73.04 75.92 77.00 77.00 78.98 80.96
## 231 62.96 68.00 68.00 69.98 69.08 75.02 69.08 62.06    NA
## 232 42.08 51.08 51.98 53.06 53.96 46.04 37.04 46.94 51.08
## 233 37.04 44.96 46.04 53.06    NA 44.96 28.94 26.06    NA
## 234 30.92 26.06 35.96 55.94 55.94 41.00 42.98 42.08 39.02
## 235 53.06 46.94 42.08 48.02 37.94 37.94 42.98 32.00 39.92
## 236 39.92 39.92 39.92 39.92 41.00 46.94    NA    NA    NA
## 237 48.92 42.08 39.92 37.94 44.96 57.92 48.02 46.04    NA
## 238 64.04 64.04 41.00 46.04 55.94 55.94 46.04 48.92    NA
## 239 82.94 82.04 78.08 78.98 80.96 78.08 73.94 71.06 69.08
## 240 84.02 78.98 77.00 77.00 77.00 78.08 78.08 82.94    NA
## 241 84.02 80.96 77.00 78.98 75.92 80.96 84.92 82.94 80.96
## 242 80.06 75.92 75.02 78.08 80.96 82.04 84.92 84.02 80.96
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## 243 75.02 73.94 71.96 75.02 77.00 73.04 71.96 71.96    NA
## 244 64.94 62.06 64.94 64.94 62.96 60.08 60.08 55.94 60.08
## 245 57.02    NA 57.92 57.92 46.04 30.92 42.98 55.04    NA
## 246 60.08 60.08 55.94 60.08 51.08 48.92 60.98 51.08 51.98
## 247 44.06 44.06 42.98 39.02 42.98 42.98    NA 48.92 57.02
## 248 46.94 50.00 55.04 57.92 59.00 60.08    NA    NA    NA
## 249 42.98 48.02 57.92 55.94 57.92 51.08 42.98 41.00 50.00
## 250 64.94 53.06 55.94 62.06 64.94 60.98 55.04 66.92    NA
## 251 62.06 73.04 80.96 78.08 73.04 66.92 62.06 66.92 73.94
## 252 86.00 89.06 89.96 89.06 80.96 78.08 78.08 80.06    NA
## 253 86.00 87.08 87.98 87.98 82.04 78.08 82.04 75.92 73.04
## 254 75.92 75.92 75.92 71.96 73.04 77.00 78.08 78.08 77.00
## 255 59.00 55.94 62.06 73.94 73.04 73.04 75.02 77.00    NA
## 256 59.00 66.92 53.96 55.04 48.92 55.94 62.06 57.92 57.02
## 257    NA    NA 60.08 48.92 46.94 39.92 33.08 28.94    NA
## 258    NA 48.92 46.04 55.04 55.94 39.92 32.00 48.02 51.98
## 259 30.92 46.04 44.06 42.08 42.98 46.04 42.98 50.00 35.96
## 260 53.06 39.02 39.92    NA 44.96 46.94    NA    NA    NA
## 261 51.08 50.00 51.08 59.00 60.08 48.92 35.06 41.00 62.06
## 262 50.00 46.04 59.00 64.04 64.94 71.96 75.92 73.04    NA
## 263 62.96 53.96 66.92 73.04 75.02 75.02 73.04 75.02 75.92
## 264    NA 87.98 82.94 84.02 86.00 84.02 84.02    NA    NA
## 265 77.00 73.04 73.04 75.02 78.08    NA 77.00 78.08 75.02
## 266 82.94 80.96 84.02 80.06 75.02 78.08 80.96 82.94 80.06
## 267 66.02 57.92 60.98 64.94 69.98 71.06 64.94 64.04    NA
## 268 53.96 64.94 71.06 71.96 69.08 68.00 69.08 69.98 62.06
## 269 69.08 42.98    NA 53.96 53.06 51.98    NA 42.98    NA
## 270 35.06 42.98 55.04 30.92 33.98 24.98 32.00 39.92 48.02
## 271 39.92 41.00 30.92 37.04 42.98 39.92 30.92 33.08 26.96
## 272 33.08 42.98 39.92 48.92 44.96 55.04 57.92    NA    NA
## 273 55.94 55.04 64.04 60.98 60.08 51.98 59.00 57.02 51.98
## 274 64.04 60.98 60.98 64.04 66.02 60.98 71.06 69.98    NA
## 275 48.02 42.08 50.00 57.02 55.04 64.94 64.04 66.02 69.08
## 276 82.94 82.94 82.04 80.06 78.08 80.96 86.00 87.08    NA
## 277 80.06 82.04 82.94 82.94 75.02 78.08 80.06 82.94 84.92
## 278 80.06 80.96 80.96 75.92 71.96 78.98 84.02 80.96 77.00
## 279 71.06 73.04 77.00 77.00 75.92 75.92 73.94 73.04    NA
## 280 53.06 62.06 69.98 75.92 71.96 73.94 71.06 71.06 68.00
## 281 57.92 59.00 53.06 50.00 42.98 41.00 44.96 53.06    NA
## 282 33.08 28.94 33.98 33.98 35.06 26.96 44.06 51.08 51.08
## 283 50.00 46.04 46.04 42.08 35.06 35.96 42.98 42.08 44.06
## 284 55.94 57.92 57.92 57.92 51.08 51.98    NA    NA    NA
## 285 39.02 51.08 51.98 46.94 44.06 57.92 53.06 48.02 55.04
## 286 66.02 62.96 53.96 55.94 59.00 62.06 66.02 68.00    NA
## 287 78.98 80.96 80.96 77.00 78.98 84.92 82.04 86.00    NA
## 288 82.94 82.04 80.96 82.94 87.08 87.98 84.02 78.98 82.94
## 289 78.98 77.00 75.92 80.06 84.02 86.00 84.02 84.92 82.94
## 290 66.92 69.98 80.06 80.06 82.94 78.98 75.92 66.02    NA
## 291 66.02 66.92 59.00 55.94 50.00 30.92 37.04 57.02 60.98
## 292 46.04 41.00 50.00 53.06 55.04 44.06 37.04 39.02    NA
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## 293 32.00 32.00 33.08 26.96 39.02 26.96 37.04 28.04 30.92
## 294 26.06 30.92 41.00 46.94 39.02 39.02 42.98 42.98 42.98
## 295 33.98 42.08 42.98 39.02 42.98 39.92    NA    NA    NA
## 296 44.96 48.92 51.08 44.06 42.08 55.94 60.98 59.00 48.92
## 297 42.08 60.08 64.04 66.92 66.02 64.04 44.06 44.96    NA
## 298 53.96 55.04 64.04 68.00 71.06    NA    NA    NA 71.96
## 299 82.04 84.92 82.94 82.94 78.98 84.02 80.96 78.98    NA
## 300 73.04 80.06 78.98 77.00 77.00 77.00 75.02 75.92 66.92
## 301 78.08 80.06 78.98 80.06 77.00 71.06 69.98 68.00 71.96
## 302 71.06 71.96 75.92 80.06 82.04 82.94 82.94 82.94    NA
## 303 51.08 55.04 51.08 51.08 48.92 60.08 64.94 60.98 59.00
## 304 35.96 35.06 28.94 41.00 46.94 30.92 32.00 39.02    NA
## 305 44.96 35.96 33.98 39.02 37.94 35.06 44.96 30.92 26.96
## 306 39.92 42.08 48.02 46.04 48.92 53.06 50.00 42.08 35.06
## 307 39.92 37.94 41.00 35.96 28.94 48.92    NA    NA    NA
## 308 48.92 46.94 44.96 44.06 53.96 55.94 57.92 64.94 69.98
## 309 57.92 57.02 62.06 55.04 59.00 66.02 60.98 48.92    NA
## 310 60.98 62.96 73.04 73.04 73.94 71.06 60.08 60.98 71.96
## 311 84.02 84.02 82.94 84.02 87.08 89.06 82.94 80.96    NA
## 312 84.92 87.08 78.98 71.96 82.94 82.94 82.04 78.08 78.98
## 313 84.92 87.08 80.96 80.06 84.02 84.02 84.92 84.92 84.02
## 314 78.98 78.08 75.02 71.06 75.92 77.00 77.00 78.98    NA
## 315 69.98 68.00 64.94 57.02 50.00 55.94 66.02 64.94 66.02
## 316 60.08 55.04 51.98 48.92 57.02 64.94 60.98 48.92    NA
## 317 32.00 39.02 44.96 51.98 59.00 57.92 59.00 57.92 53.06
## 318 37.94 39.92 53.96 53.96 42.98 42.08 51.08 48.92 46.94
## 319 55.04 46.94 53.96 51.08 46.94 35.06 44.06    NA    NA
## 320 57.02 57.92 55.94 53.06 59.00 60.98    NA    NA    NA
## 321 75.92 71.06 69.08 53.96 62.06 64.94 66.02 68.00    NA
## 322 71.96 64.94 53.96 55.04 62.96 71.06 73.94 78.08 82.04
## 323 82.94 82.94 82.04 82.04 82.04 84.92 86.00 87.08    NA
## 324 77.00 73.04 80.06 80.96 78.98 80.96 78.08 75.02 77.00
## 325 69.98 71.96 78.08 73.94 78.08 78.98 80.06 77.00 77.00
## 326 78.08 71.96 71.06 69.08    NA    NA    NA 73.94    NA
## 327 60.08 57.92 55.04    NA 62.96 69.08 68.00 68.00 66.92
## 328 57.92 62.96 57.92 55.94 55.94 55.94 53.96 55.04    NA
## 329 42.98 32.00 32.00 30.02 33.08 35.96 35.06 30.02 30.02
## 330 51.08 51.08 48.92 48.02 39.02 28.04 42.98 48.92 53.06
## 331 39.02 44.96 44.06 37.04    NA 44.96    NA    NA    NA
## 332 51.08 57.02 57.02 57.92 57.92 62.96 62.06 62.06 62.06
## 333 62.96 62.06 62.06 69.98 73.04 75.02 73.94 69.98    NA
## 334 69.98 71.06 71.06 69.98 69.98 71.06 73.94 78.08 78.98
## 335 77.00 78.98 82.94 91.04 93.92 89.06 87.98 89.06    NA
## 336 78.98 77.00 73.04 75.92 75.02 75.02 80.06 82.94 82.04
## 337 75.92 69.08 62.06 69.98 71.96 75.02 75.02 71.96    NA
## 338 71.06 69.98 66.02 62.06 57.02 64.04 69.08 69.98    NA
## 339 66.02 62.96 60.08 60.08 60.98 51.98 51.08 42.98 53.06
## 340 33.98 33.98 46.04 42.98 39.92 35.96 46.94 48.02    NA
## 341    NA    NA    NA    NA    NA    NA 35.06    NA    NA
## 342 50.00 39.92 55.94 55.04 51.98 46.04 55.04 51.08 41.00
```

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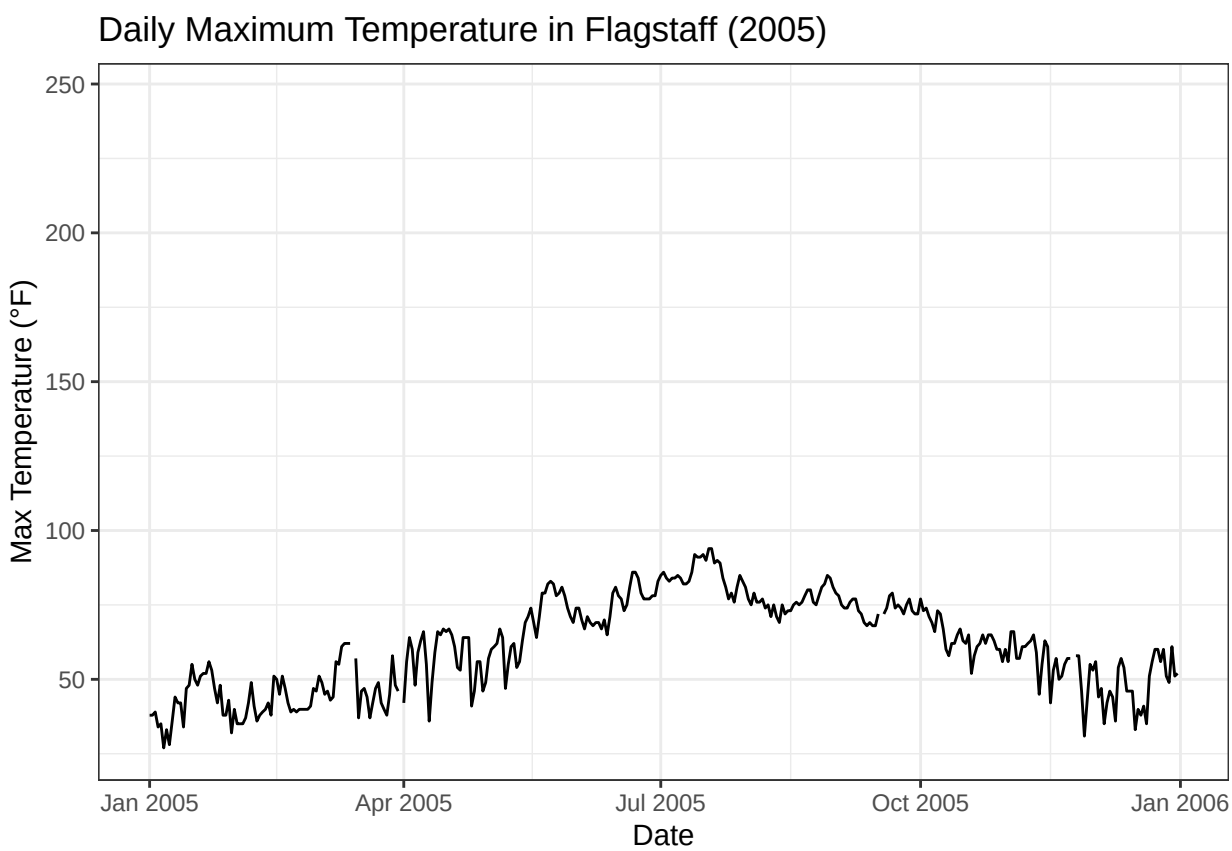
## 343    NA 57.92 55.94 53.06    NA    NA    NA    NA    NA
## 344    NA    NA    NA    NA    NA    NA    NA    NA    NA
## 345 55.94 64.04 62.06 60.08 44.06 46.94 46.94 46.04    NA
## 346 60.98 59.00 69.98 77.00 78.98 78.98 75.02 77.00 78.08
## 347 80.06 80.06 80.06 78.98 78.98 84.02 84.92 86.00    NA
## 348 84.92 84.02 84.92 84.92 87.98 73.04 80.06 82.94 80.96
## 349 69.08 71.96 71.06 66.02 71.06 73.94 78.08 78.08 77.00
## 350 73.94 77.00 75.92 73.94 60.08 57.92 60.08 64.94    NA
## 351 69.98 71.96 71.96 66.02 66.02 66.02 66.92 69.08 66.02
## 352 50.00 50.00 51.08 62.06 64.94 64.94 55.94 51.08    NA
## 353 51.08 53.06 32.00 26.96 28.94 39.92 39.92 42.08 21.02
## 354 46.04 46.04 53.06 53.96    NA 53.96    NA    NA 46.04
## 355 35.96 35.06 46.94 44.96 42.08 39.92    NA    NA    NA
## 356 60.08 59.00 62.06 60.08 69.98 69.08 68.00 68.00 66.92
## 357 57.02 53.96 46.04 48.92 57.92 69.08 68.00 71.06    NA
## 358 53.96 55.94 57.92 66.02 66.92 71.96 71.96 77.00 78.08
## 359 84.92 86.00 84.92 89.06    NA 80.96 82.94 87.08    NA
## 360 77.00 78.08 80.06 80.06 80.06 82.04 80.96 73.04 78.08
## 361 78.08 82.04 73.94 73.04    NA 77.00 82.94 80.96 75.02
## 362 75.92 77.00 77.00 75.92 77.00 75.92 78.98 78.98    NA
## 363 57.92 59.00 60.98 62.96 64.94 62.06 60.08 51.98 60.98
## 364 59.00 55.94 48.92 42.08 33.08 37.94 35.96 42.98    NA
## 365    NA    NA 35.06 23.00 37.94    NA    NA    NA    NA
```

```r
MaxTemp_long <- MaxTemp %>%
  pivot_longer(
    cols = starts_with("X"),
    names_to = "Day",
    values_to = "MaxTemp"
  ) %>%
  mutate(
    Day = as.numeric(gsub("X", "", Day)),
    Date = make_date(Year, Month, Day)
  )
MaxTemp_2005 <- MaxTemp_long %>%
  filter(Year == 2005)
ggplot(MaxTemp_2005, aes(x = Date, y = MaxTemp)) +
  geom_line() +
  labs(
    title = "Daily Maximum Temperature in Flagstaff (2005)",
    x = "Date",
    y = "Max Temperature (°F)"
  ) +
  theme_bw()
```

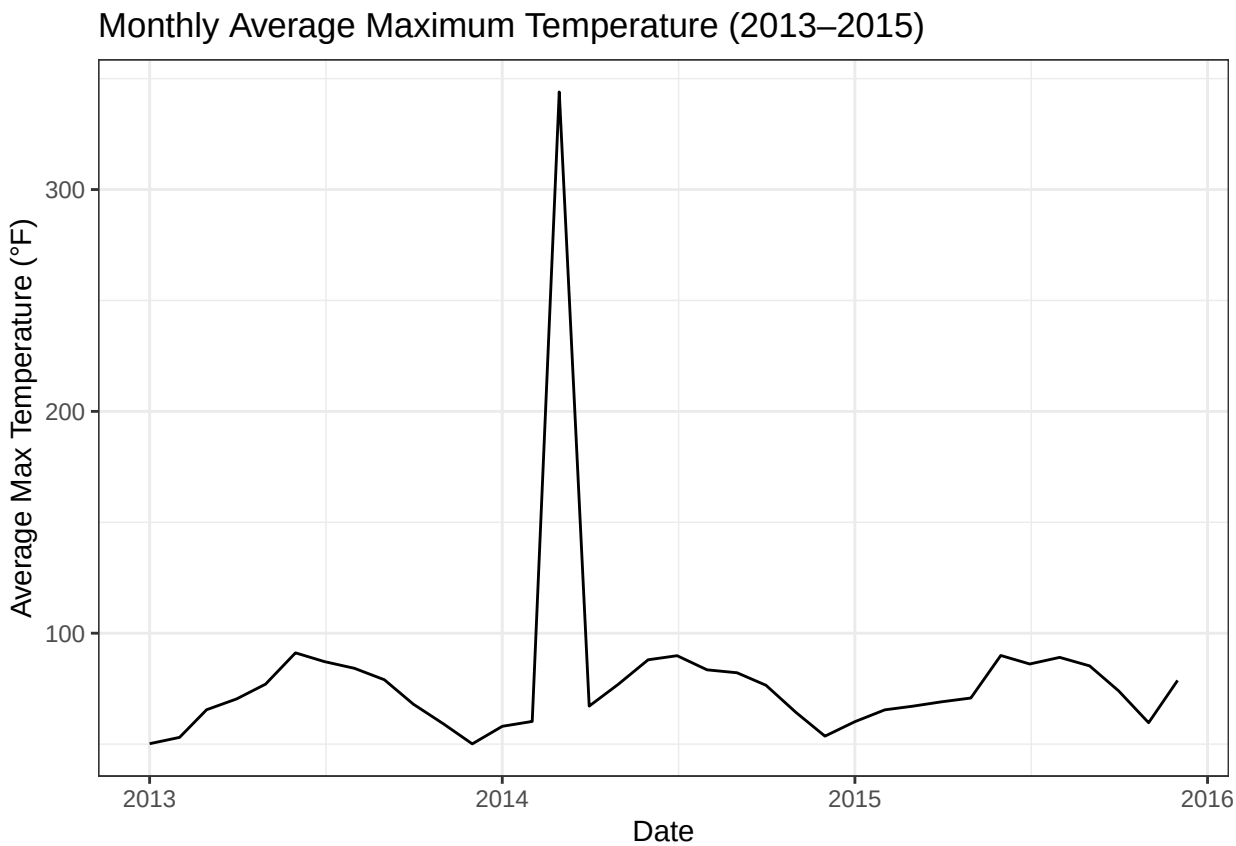
```
## Warning: Removed 19 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



b) Create a line graph that gives the monthly average maximum temperature for 2013 - 2015. *Again the x-axis should be the date and span 3 years.*

```
MaxTemp_long <- MaxTemp %>%
  pivot_longer(
    cols = starts_with("X"),
    names_to = "Day",
    values_to = "MaxTemp"
  ) %>%
  mutate(
    Day = as.numeric(gsub("X", "", Day)),
    Date = make_date(Year, Month, Day)
  ) %>%
  filter(!is.na(MaxTemp))
MaxTemp_monthly <- MaxTemp_long %>%
  filter(Year >= 2013 & Year <= 2015) %>%
  group_by(Year, Month) %>%
  summarise(
    AvgMaxTemp = mean(MaxTemp, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(Date = make_date(Year, Month, 1))
ggplot(MaxTemp_monthly, aes(x = Date, y = AvgMaxTemp)) +
  geom_line() +
  labs(
```

```
    title = "Monthly Average Maximum Temperature (2013-2015)",
    x = "Date",
    y = "Average Max Temperature (°F)"
  ) +
  theme_bw()
```



Problem 4

The package `nycflights13` contains information about all the flights that arrived in or left from New York City in 2013. This package contains five data tables, but there are three data tables we will work with. The data table `flights` gives information about a particular flight, `airports` gives information about a particular airport, and `airlines` gives information about each airline. Create a table of all the flights on February 14th by Virgin America that has columns for the carrier, destination, departure time, and flight duration. Join this table with the airports information for the destination. Notice that because the column for the destination airport code doesn't match up between `flights` and `airports`, you'll have to use the `by=c("TableA.Col"="TableB.Col")` argument where you insert the correct names for `TableA.Col` and `TableB.Col`.

```
va_flights <- flights %>%
  filter(month == 2, day == 14, carrier == "VX") %>%   # Virgin America = "VX"
  select(carrier, dest, dep_time, air_time)
va_flights_joined <- va_flights %>%
  left_join(airports, by = c("dest" = "faa"))
va_flights_joined
```

```
## # A tibble: 10 x 11
##    carrier dest  dep_time air_time name        lat   lon   alt    tz dst   tzone
##    <chr>   <chr>    <int>    <dbl> <chr>     <dbl> <dbl> <dbl> <dbl> <chr> <chr>
##  1 VX      LAX        706      347 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  2 VX      SFO        732      344 San Fran~  37.6 -122.    13    -8 A     Amer~
##  3 VX      LAX        909      341 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  4 VX      LAS        934      307 Mc Carra~  36.1 -115.  2141    -8 A     Amer~
##  5 VX      SFO       1029      351 San Fran~  37.6 -122.    13    -8 A     Amer~
##  6 VX      LAX       1317      349 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  7 VX      LAX       1706      335 Los Ange~  33.9 -118.   126    -8 A     Amer~
##  8 VX      SFO       1746      358 San Fran~  37.6 -122.    13    -8 A     Amer~
##  9 VX      SFO       1852      355 San Fran~  37.6 -122.    13    -8 A     Amer~
## 10 VX      LAX       2017      337 Los Ange~  33.9 -118.   126    -8 A     Amer~
```

Problem 5

Data table joins are extremely common because effective database design almost always involves having multiple tables for different types of objects. To illustrate both table joins and the usefulness of multiple tables we will develop a set of data frames that will represent a credit card company's customer data base. We will have tables for Customers, Retailers, Cards, and Transactions. Below is code that will create and populate these tables.

```r
Customers <- tribble(
  ~PersonID, ~Name, ~Street, ~City, ~State,
  1, 'Derek Sonderegger',  '231 River Run', 'Flagstaff', 'AZ',
  2, 'Aubrey Sonderegger', '231 River Run', 'Flagstaff', 'AZ',
  3, 'Robert Buscaglia', '754 Forest Heights', 'Flagstaff', 'AZ',
  4, 'Roy St Laurent', '845 Elk View', 'Flagstaff', 'AZ')

Retailers <- tribble(
  ~RetailID, ~Name, ~Street, ~City, ~State,
  1, 'Kickstand Kafe', '719 N Humphreys St', 'Flagstaff', 'AZ',
  2, 'MartAnnes', '112 E Route 66', 'Flagstaff', 'AZ',
  3, 'REI', '323 S Windsor Ln', 'Flagstaff', 'AZ' )

Cards <- tribble(
  ~CardID, ~PersonID, ~Issue_DateTime, ~Exp_DateTime,
  '9876768717278723',  1,  '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '5628927579821287',  2,  '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '7295825498122734',  3,  '2019-9-28 0:00:00', '2022-9-28 0:00:00',
  '8723768965231926',  4,  '2019-9-30 0:00:00', '2022-9-30 0:00:00' )

Transactions <- tribble(
  ~CardID, ~RetailID, ~DateTime, ~Amount,
  '9876768717278723', 1, '2019-10-1 8:31:23',    5.68,
  '7295825498122734', 2, '2019-10-1 12:45:45',  25.67,
  '9876768717278723', 1, '2019-10-2 8:26:31',    5.68,
  '9876768717278723', 1, '2019-10-2 8:30:09',    9.23,
  '5628927579821287', 3, '2019-10-5 18:58:57',  68.54,
```

```
  '7295825498122734', 2, '2019-10-5 12:39:26',  31.84,
  '8723768965231926', 2, '2019-10-10 19:02:20', 42.83)

Cards <- Cards %>%
  mutate( Issue_DateTime = lubridate::ymd_hms(Issue_DateTime),
          Exp_DateTime   = lubridate::ymd_hms(Exp_DateTime) )
Transactions <- Transactions %>%
  mutate( DateTime = lubridate::ymd_hms(DateTime))
```

a) Create a table that gives the credit card statement for Derek. It should give all the transactions, the amounts, and the store name. Write your code as if the only initial information you have is the customer's name. *Hint: Do a bunch of table joins, and then filter for the desired customer name. To be efficient, do the filtering first and then do the table joins.*

b) Aubrey has lost her credit card on Oct 15, 2019. Close her credit card at 4:28:21 PM and issue her a new credit card in the `Cards` table. *Hint: Using the Aubrey's name, get necessary CardID and PersonID and save those as `cardID` and `personID`. Then update the `Cards` table row that corresponds to the `cardID` so that the expiration date is set to the time that the card is closed. Then insert a new row with the `personID` for Aubrey and a new `CardID` number that you make up.*

c) Aubrey is using her new card at Kickstand Kafe on Oct 16, 2019 at 2:30:21 PM for coffee with a charge of $4.98. Generate a new transaction for this action. *Hint: create temporary variables `card`,`retailid`,`datetime`, and `amount` that contain the information for this transaction and then write your code to use those. This way in the next question you can just use the same code but modify the temporary variables. Alternatively, you could write a function that takes in these four values and manipulates the tables in the GLOBAL environment using the `<<-` command to assign a result to a variable defined in the global environment. The reason this is OK is that in a real situation, these data would be stored in a database and we would expect the function to update that database.*

d) On Oct 17, 2019, some nefarious person is trying to use her OLD credit card at REI. Make sure your code in part (c) first checks to see if the credit card is active before creating a new transaction. Using the same code, verify that the nefarious transaction at REI is denied. *Hint: your check ought to look something like this:*

```
card <- '9876768717278723'
retailid <- 2
datetime <- ymd_hms('2019-10-16 14:30:21')
amount <- 4.98

# If the card is currently valid, this should return exactly 1 row.
Valid_Cards <- Cards %>%
  filter(CardID == card, Issue_DateTime <= datetime, datetime <= Exp_DateTime)

# If the transaction is valid, insert the transaction into the table
if( nrow(Valid_Cards) == 1){
  # Some code to insert the transaction
}else{
  print('Card Denied')
}
```

e) Generate a table that gives the credit card statement for Aubrey. It should give all the transactions, amounts, and retailer name for both credit cards she had during this period.